applied sciences

MDPI

Review

Agricultural AI Agents: Architecture Design, Business Processes, Key Technologies, and Future Challenges

Xuehua Song, Li Han, Yi Zhu, Qianxiang Wei, Zijun Yang and Xiaoming Jiang *

School of Computer Science and Computer Science and Communication Engineering, Jiangsu University, Zhenjiang 212013, China; songxh@ujs.edu.cn (X.S.); hanli_study@stmail.ujs.edu.cn (L.H.); zhuyi@ujs.edu.cn (Y.Z.); wqxtbs0213@163.com (Q.W.); 2222508020@stmail.ujs.edu.cn (Z.Y.)

* Correspondence: jxm@ujs.edu.cn

Abstract

Agricultural AI agents play a crucial role in the evolution of smart agriculture, from single-point automated applications to intelligent systems driven by tasks, collaborative decision-making, and closed-loop execution. However, their practical implementation still faces key challenges, such as heterogeneous agricultural data processing, insufficient cross-scenario generalization ability, complexity of multi-agent collaboration, difficulties in integrating software and hardware, and insufficient security and trust guarantees in real agricultural environments. This paper presents a systematic review of the architecture design, business processes, key technologies, and future challenges of agricultural AI agents. Agricultural AI agents are classified into two types: virtual agricultural AI agents and embodied agricultural AI agents. The paper summarizes a four-layer system architecture consisting of the infrastructure layer, agent management layer, agent collaboration layer, and application layer. The paper also analyzes the model capabilities required by agricultural AI agents from four typical business dimensions: perception and state understanding, knowledge memory and experience management, reasoning decision-making and task planning, and collaborative execution and resource scheduling. This research shows that technologies such as multimodal perception, knowledge graphs, retrieval-enhanced generation, digital twins, reinforcement learning, and multi-agent collaboration can provide important support for agricultural AI agents to enhance their environmental understanding, knowledge reuse, autonomous decision-making, and physical execution capabilities. Future research should focus on robust perception in open environments, long-term memory and knowledge evolution, reliable multi-agent collaboration, edge-cloud collaborative deployment, and secure and trustworthy human–machine collaboration. Integrating agricultural domain knowledge with intelligent agent technology is an important direction for promoting the large-scale, adaptive, and sustainable application of agricultural AI agents.

Keywords: agricultural AI agents; smart agriculture; multi-agent systems; task planning; collaborative execution

Check for updates

Academic Editor: José Miguel Molina Martínez

Received: 9 May 2026
Revised: 21 May 2026
Accepted: 23 May 2026
Published: 28 May 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

1.1. Research Background: From Smart Agriculture to Agricultural AI Agents

Global agricultural production is shifting from experience- and mechanization-driven modes toward data- and intelligence-driven paradigms. Crop production exhibits significant spatiotemporal heterogeneity as soil conditions, moisture, meteorology, crop growth, pest and disease outbreaks, and agricultural machinery status vary across regions, seasons,

crop varieties, and management practices. Although traditional agricultural information systems can support data acquisition, remote monitoring, and localized automatic control, they remain limited in multi-source data fusion, cross-task collaborative decision-making, swarm equipment coordination, and closed-loop feedback. In recent years, artificial intelligence has been widely applied in smart agriculture, covering crop recognition, agricultural machinery navigation, pest and disease detection, smart irrigation, greenhouse control, and agricultural equipment automation, thereby gradually forming an intelligent technology system for the entire agricultural production process.

Existing reviews show that AI has played an important role in planting, management, harvesting, agricultural machinery, agricultural IoT, and food safety, but it still faces challenges such as distorted data quality, weak adaptability to complex environments, limited model generalization, and difficulties in heterogeneous software–hardware collaboration [1]. Research on agricultural equipment intelligence further indicates that the integration of AI models with multi-modal perception, navigation control, and precision operation technologies is improving equipment efficiency, resource utilization, and environmental adaptability [2]. From the perspectives of autonomous navigation, operational control, power systems, and equipment intelligence, agricultural machinery has gradually evolved from mechanical execution tools into intelligent carriers capable of autonomous perception, reasoning, and collaborative operation [3]. The existing literature suggests that the key bottleneck of smart agriculture is no longer simply whether systems can perceive or control automatically but how to organize perception, reasoning, planning, execution, and feedback into a continuous agricultural task chain.

At the perception level, information on crop pests and diseases, weeds, soil moisture, crop canopy, and machinery status forms the foundation of agricultural decision-making. For example, the YOLOv8-GDCI model for detecting different infected parts of pepper blight shows that complex backgrounds, occlusions, small targets, and multi-scale lesion differences strongly affect the robustness and practical value of agricultural vision models [4]. At the equipment level, boom height detection based on ultrasonic sensors provides canopy height information for adaptive spraying control during wheat growth [5], while further optimization considers the influence of boom oscillation on detection results [6]. Existing research indicates that intelligence in real agricultural environments is not determined by a single model's recognition accuracy but by the collaboration among sensors, scene constraints, equipment execution, and control objectives.

1.2. Proposal of the Agricultural AI Agent Concept

Against this background, an agricultural AI agent can be understood as a highly autonomous intelligent unit oriented toward agricultural production, management, and service tasks. It can perceive agricultural environments, invoke knowledge and tools, conduct task planning, and participate in decision-making or execution. In this review, the term "agricultural AI agent" is used as a unified expression. It refers to an autonomous or semi-autonomous AI-enabled entity that can perceive agricultural states, invoke models or tools, reason about domain knowledge, plan tasks, and participate in decision-making or physical execution. The terms "agricultural intelligent agent" and "agricultural agent" are used only when referring to terminology adopted in cited studies, but they are treated as conceptually equivalent to agricultural AI agents in the context of this review. Similarly, the term "embodied AI agent" is used to describe agricultural AI agents with physical carriers or direct interaction capabilities in real agricultural environments, while "embodied intelligent agent" is treated as a conceptually equivalent expression when it appears in the cited literature. Compared with traditional smart agriculture systems, agricultural AI agents emphasize task-driven evolution, heterogeneous tool invocation, distributed

collaboration, and full-chain closed-loop feedback. They may exist as virtual agents running on cloud, edge, or terminal devices for data processing, multi-modal knowledge retrieval, scheme generation, and task scheduling or as embodied agents with physical carriers, such as agricultural robots, drones, automatic sprayers, smart irrigation equipment, and greenhouse control systems, for real agricultural operations.

The emergence of agricultural AI agents is closely related to the development of foundation models, digital twins, reinforcement learning, and intelligent equipment. Foundation models are expected to improve the adaptability of agricultural models across tasks and scenarios through pre-training and transfer learning, thereby providing general cognitive and reasoning support for agricultural agents [7]. Digital twins can support crop management, pest and disease prevention, machinery optimization, and decision support through real-time data, simulation models, and virtual–real mapping [8]. Deep reinforcement learning provides technical support for dynamic decision-making tasks such as irrigation scheduling, path planning, robot control, and resource allocation [9]. Together, these technologies constitute the foundation of agricultural AI agents: foundation models serve as knowledge hubs, digital twins provide state synchronization and virtual verification, and reinforcement learning and control algorithms support dynamic decision-making and execution optimization.

Meanwhile, agent technology offers a new organizational paradigm for integrating these decentralized technologies. The Agentic AI framework designs soil monitoring, weather perception, disease detection, and supervisory chatbots as agricultural agents with different roles, demonstrating the basic form of multi-agent collaboration in climate-smart agriculture [10]. The "Fuxi Brain" further presents an agricultural autonomous decision-making pathway from data collection to full-production-cycle decision-making through a "space–air–ground–human–machine" data acquisition system, generative large models, and a multi-agent collaborative architecture [11]. Therefore, agricultural AI agents are not simple replacements for traditional AI models but a new form of agricultural intelligent system that organizes data, models, knowledge, tools, and equipment.

1.3. Essential Differences Between Agricultural AI Agents and Traditional Systems

The essential difference between agricultural AI agents and traditional smart agriculture systems lies in task organization. More specifically, AI-enabled agricultural systems mainly use AI models as embedded functional modules for recognition, prediction, or control, while agricultural AI agents emphasize autonomous task organization. An AI-enabled irrigation controller, for example, may predict soil moisture and open valves according to predefined rules; an agricultural AI agent should further interpret the irrigation goal, retrieve contextual knowledge, select appropriate tools, coordinate sensing and execution devices, monitor the operation result, and update the task record after execution. Therefore, the boundary between the two lies not in whether AI models are used but in whether perception, reasoning, planning, execution, and feedback are organized into an autonomous task loop. Traditional systems usually focus on single functions, such as disease recognition, irrigation control, spraying, or path planning. In contrast, agricultural AI agents emphasize full-link task flows, covering anomaly discovery, task decomposition, agent scheduling, scheme generation, equipment execution, and result feedback. Reviews of control algorithms show that smart agriculture control has expanded from PID and fuzzy control to neural networks, deep learning, reinforcement learning, and multi-objective optimization [12]. These methods can solve local control problems, but only when embedded into task-driven agent workflows can they form system-level collaborative capabilities.

Agricultural AI agents also require stronger multi-modal perception and cross-scenario generalization. Reviews of agricultural pest and disease detection indicate that deep learn-

ing is driving pest monitoring from manual inspection toward automation, real-time monitoring, and precision [13]. Environmental perception studies for orchard targeted spraying further show that orchard operations require the integration of canopy structure, wind speed and direction, spatial location, target distribution, and spray parameters [14]. These findings suggest that agricultural AI agents must cope with more complex physical environments than general internet agents, and their decisions depend not only on language and knowledge but also on images, sensors, equipment status, and agricultural mechanistic models.

In addition, agricultural AI agents require deep synergy between digital decision-making and physical execution. A pure agricultural Q&A system can provide suggestions, but changing agricultural production requires translating suggestions into control commands for machinery operation, spraying, irrigation, ventilation, fertilization, or harvesting. Research on agricultural equipment intelligence points out that future agricultural systems will increasingly integrate digital twins, edge computing, and big-data collaborative optimization with equipment operations [2]. Therefore, agricultural AI agents must combine high-level cognitive decision-making with low-level execution interfaces through communication protocols, edge computing, task offloading, and equipment control.

1.4. Key Challenges Currently Facing Agricultural AI Agents

Although agricultural AI agents provide a new organizational paradigm for smart agriculture, their engineering implementation still faces several challenges. The first is heterogeneous data processing and semantic alignment. Agricultural data come from remote sensing images, ground sensors, machinery controllers, weather stations, operational records, and farmer experience. Their heterogeneity, high annotation cost, and inconsistent spatiotemporal scales affect agent perception and reasoning. Digital twin research also points out that data acquisition, model integration, and 3D crop model standardization remain major obstacles [8].

The second challenge is model generalization and domain adaptation. Agricultural scenes vary greatly in crops, regions, lighting, soil, cultivation methods, and equipment conditions, making it difficult for models trained on experimental datasets to transfer directly to production environments. Pepper blight detection requires datasets and model structures adapted to infected leaves, fruits, and stems [4], while boom height detection must consider wheat growth stages and equipment oscillation [5,6]. Existing research shows that the generalization ability of agricultural AI agents is jointly constrained by models, data, sensors, and physical operation conditions.

The third challenge is multi-agent collaboration and swarm intelligence. Agricultural production involves multiple entities, including farmers, agronomists, management platforms, sensing devices, machinery, and market services. A review of multi-agent large language models indicates that agricultural multi-agent systems need to address role division, communication protocols, task decomposition, result integration, and safety and trustworthiness, forming a collaborative chain of data collection, processing, task coordination, execution, and feedback optimization [15].

The fourth challenge is software–hardware co-design. Agricultural AI agents must connect cloud models, edge reasoning, sensor networks, and physical equipment, which imposes high requirements on real-time performance, reliability, energy consumption, and interface standards. The fifth challenge is safety and trustworthiness. Incorrect decisions may cause pesticide misuse, equipment collisions, water and fertilizer waste, or crop losses. Therefore, interpretability, decision traceability, human–machine supervision, and safety constraints are essential for ensuring compliant and controllable agent behavior in complex agricultural ecosystems.

1.5. Review Scope and Methodological Framework

Based on the above background, this paper reviews the conceptual connotations, classification systems, system architectures, typical workflows, key technologies, and future challenges of agricultural AI agents. Rather than proposing a new algorithm, this review aims to clarify how agricultural AI agents are defined, organized, and applied in real agricultural scenarios. Specifically, this paper focuses on three core questions. First, what are the essential connotations and architectural characteristics of agricultural AI agents? Second, how are agricultural AI agent systems organized and operated, especially in terms of virtual agents, embodied agents, and multi-agent collaboration? Third, how do agricultural AI agents realize the full-chain closed loop from perception and decision-making to execution and feedback in real agricultural tasks?

The literature analysis in this study consisted of two stages: first, literature retrieval and bibliometric analysis were conducted; second, representative studies were selected for in-depth review. Relevant publications were mainly collected from databases such as Web of Science, ScienceDirect, IEEE Xplore, and Google Scholar. The search terms were constructed around agricultural application scenarios, agricultural AI agent technologies, and enabling AI methods, including ["agricultural AI agent"] AND ["multi-agent system AND agriculture"]. The temporal scope of the literature search covered studies published from 2020 to 2026, with a focus on large models, multi-agent collaboration, embodied intelligence, edge–cloud deployment, and agricultural decision support.

To ensure relevance and reproducibility, explicit inclusion and exclusion criteria were applied. Studies were included if they were related to agricultural AI agents, agricultural intelligent systems, agent-based modeling, multi-agent collaboration, or autonomous agricultural decision-making and if they addressed key capabilities, such as perception, reasoning, planning, collaboration, execution, or feedback-based optimization. Studies were excluded if they were unrelated to agricultural scenarios, focused only on general AI algorithms without agricultural applications or agent-related content, were duplicate or non-peer-reviewed records, or lacked sufficient methodological information. The final cited works were selected through title and abstract screening, full-text reading, and relevance evaluation according to the thematic structure of this review.

To further understand the thematic distribution of agricultural AI agent research, this study used VOSviewer software, version 1.6.20, to conduct a keyword co-occurrence analysis of the selected literature. In Figure 1, node size represents keyword frequency, links represent co-occurrence relationships, and colors represent different research clusters. Keywords such as agent-based model, agent-based modeling, agriculture, land use, ecosystem services, and multi-agent systems are located at the core of the network, indicating that current research mainly focuses on agent-based methods, agricultural system analysis, resource management, and multi-agent collaboration.

The clustering results reveal several major research directions. The green cluster focuses on agent-based model, agent-based modeling, systems, framework, behavior, and optimization, reflecting applications in agricultural system modeling, behavior analysis, framework construction, and optimization decision-making. The red cluster is associated with simulation, adaptation, climate change, land use change, biodiversity, and environment, indicating the use of agricultural AI agent methods to analyze responses to climate change, land use change, and environmental pressure. The blue cluster involves ecosystem services, land use, food security, and agricultural policy, showing the role of agricultural AI agents in ecosystem service assessment, food security analysis, and policy research. The yellow cluster involves adoption, diffusion, technology adoption, risk, allocation, and groundwater, mainly reflecting agricultural technology adoption, risk decision-making, and resource allocation. The purple cluster, represented by multi-agent systems, consen-

sus, synchronization, stability, and tracking control, more directly reflects research on multi-agent collaborative control, interaction mechanisms, and complex system stability.

Based on the above bibliometric analysis, the second stage further analyzed the selected studies from five aspects: the architectural framework of agricultural AI agents; the key capabilities required by agricultural AI agents in perception, reasoning, planning, collaboration, execution, and feedback; the ways in which existing architectures and workflows support virtual agents, embodied agents, and multi-agent collaboration; the advantages and limitations of current technologies in goal understanding and agent selection, multi-agent collaboration and consensus mechanisms, and multi-agent scheduling and workflow orchestration; and the major challenges faced by future agricultural AI agent systems. These analytical perspectives also constitute the organizational logic of the following sections.

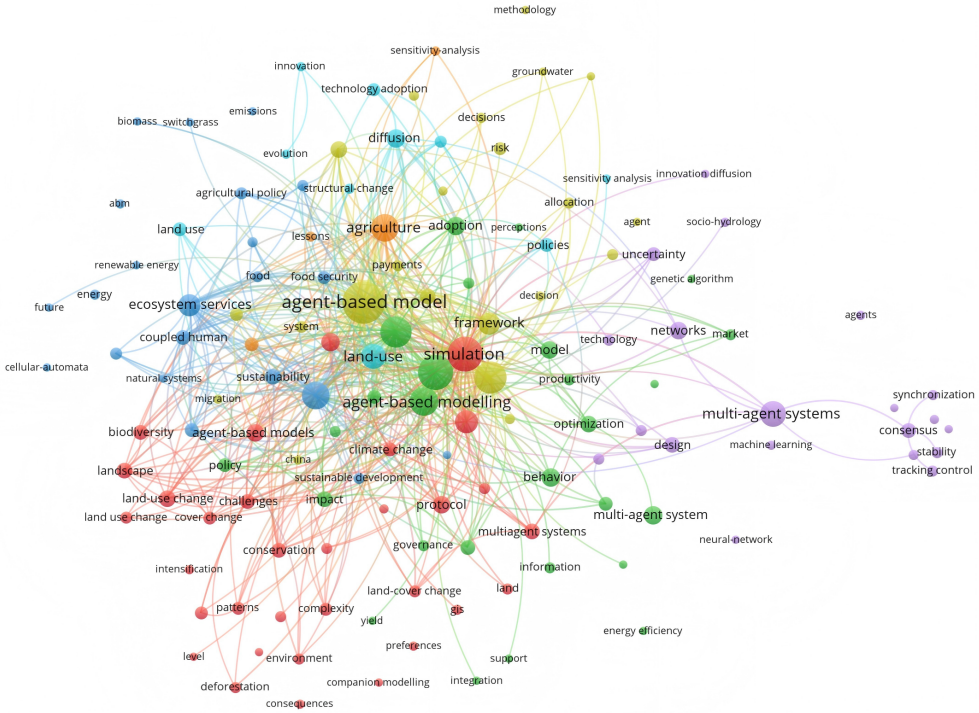


Figure 1. Keyword co-occurrence network of agricultural AI agents.

The structure of this review is organized as follows. The first part introduces the research background, concept, key issues, and review scope of agricultural AI agents. The second part discusses the classification system and system architecture, including virtual agricultural agents, embodied agricultural agents, and the four-layer architecture of infrastructure, agent management, agent collaboration, and application. The third part focuses on agricultural AI agent workflows and analyzes task triggering, agent selection, task decomposition, collaborative execution, and knowledge updating. The fourth part summarizes key technologies, including goal understanding, agent selection, multi-agent collaboration, communication protocols, workflow orchestration, and resource scheduling. The fifth part discusses future challenges, including data extraction, model generalization, multi-agent collaboration, software–hardware co-design, and safety and trustworthiness. Through this structure, this paper presents the development status and frontier landscape of agricultural AI agents along the logical chain of "Concept—Architecture—Workflow—Technology—Challenge."

2. Overview of Agricultural AI Agents: Classification System and System Architecture

Agricultural AI agents, as the product of the deep coupling of artificial intelligence technology and the agricultural production system, are fundamentally reshaping the production methods and management paradigms of traditional agriculture. To systematically grasp the essence, technology composition, and operation mechanism of agricultural AI agents, this chapter conducts progressive discussions from the perspectives of classification system and system architecture. The logical relationship between the two is as follows: the classification system answers the question, "What are the existing forms of agricultural AI agents?", while agricultural AI agent architecture answers the question, "How do agricultural AI agents organize and operate?". In simple terms, classification is the depiction of the morphological dimension of the agent, and architecture is the description of its structural dimension. They are both independent yet mutually supportive, jointly constituting the core cognitive framework of agricultural AI agents.

2.1. Classification of Agricultural AI Agents

According to whether they have physical carriers, agricultural AI agents can be classified into two basic forms: virtual agricultural AI agents and embodied agricultural AI agents. This classification reflects not only their different forms of existence but also their distinct roles in agricultural AI agent systems. Virtual agents are mainly responsible for information processing and decision-making, whereas embodied agents undertake physical operations and execution. Through cross-domain collaboration and information exchange, these two types of agents jointly constitute a complete agricultural AI-agent system.

2.1.1. Virtual Agricultural AI Agents

Virtual agricultural AI agents are software entities that operate entirely in digital environments without physical carriers. They access large models and remote knowledge bases through network connections and interact with users through graphical interfaces, voice interaction, or wearable devices. Their core feature is "invisibility": they reside in cloud servers, edge computing nodes, or terminal devices and function through information processing and decision-making rather than direct contact with the physical environment. In agriculture, these agents are becoming important bridges between data and decision-making, as well as between farmers' needs and technological services.

1. Processing and analyzing massive agricultural data
   Intelligent decision-making begins with data. Agricultural production involves multi-source heterogeneous data related to soil, weather, crops, and field operations. Multimodal sensor fusion, environmental perception, and crop status monitoring technologies provide an important foundation for agricultural data collection [14,16,17]. At the data collection stage, Goodrich et al. proposed a sequential gap reduction algorithm to optimize soil sensor placement in fields. They also used genetic algorithms to optimize the flight scanning paths of multi-agent unmanned aerial vehicles, supporting spatial data collection in large-scale farmland [18]. Betalo et al. developed a task scheduling and resource sharing framework based on multi-agent deep reinforcement learning for data collection and processing in multi-UAV-assisted wireless sensor networks, improving the efficiency of large-scale sensor–node data collection [19]. At the data management stage, Zou and Quan proposed a grid-based resource management and scheduling strategy for AIoT, supporting the unified management and task scheduling of heterogeneous agricultural data resources [20]. Gonzalez-Briones et al. developed a cloud-based multi-agent system that uses wireless sensor networks to collect potato crop data for knowledge discovery and decision-making, thereby

supporting precision irrigation. This work verifies the technical feasibility of virtual agents in agricultural data processing and engineering applications [21].

2. Data–knowledge dual-driven agricultural decision-making mechanism
Data acquisition is only the first step; the core capability of agricultural AI agents lies in transforming data into executable decisions. Agyeman et al. developed an irrigation scheduler that combines machine learning, model predictive control, and multi-agent principles, achieving 7–23% water savings and improving irrigation efficiency by 10–35% [22]. For knowledge-driven decision-making, Ge et al. proposed a recommendation model integrating knowledge graphs and case-based reasoning to predict fertilization amounts with strong interpretability and transferability [23]. Lin et al. further developed a spatiotemporal knowledge-graph-based reasoning method for rice fertilization decisions [24]. Recent advances in large language models and intelligent decision-making technologies are accelerating the transition from rule-driven to model-driven agricultural decision-making [1,25]. For example, Chen et al. developed the "Fu Xi Brain" autonomous agricultural intelligence system by integrating agricultural IoT and generative large models, improving efficiency through dynamic optimization and full-modal alignment training [11]. Collectively, the existing literature illustrates the evolution of agricultural AI agents from data perception and knowledge understanding to autonomous decision-making.
3. Providing intuitive information services and decision suggestions
Even highly accurate decisions may have limited practical value if users cannot understand or apply them. Virtual AI agents convert complex analytical results into farmer-friendly suggestions through natural language interaction. Zhang et al. developed Chat Demeter, which receives leaf images and returns disease diagnosis results and treatment suggestions with an accuracy of 99.50% [26]. Anand et al. proposed the SEEDS system, which integrates RAG, knowledge graphs, and OpenAI embedding models to provide precise pest and disease control suggestions [27]. Owiti and Kipkebut further developed a RAG-based intelligent agent with quantized fine-tuned language models for real-time, low-cost, and multilingual agricultural advice in resource-constrained areas [28]. Jayarathna and Hettige developed a MaSMT-based multi-agent communication system that supports the shift from question-answering services to intelligent consultation through role collaboration and knowledge enhancement [29]. The existing literature demonstrates that virtual AI agents serve not only as decision generators but also as semantic interfaces connecting agricultural knowledge with end farmers.
4. Cross-domain resource scheduling and multi-machine collaborative control
After decisions are generated, multiple physical devices often need to collaborate during execution, with virtual agricultural AI agents serving as the "coordination center". Skobelev et al. proposed a unified multi-agent software and information environment to support crop variety selection, rotation planning, agricultural technology formulation, and adaptive scheduling of machinery and material resources [30]. To address fuzzy time windows and matching constraints in heterogeneous agricultural robot scheduling, Guo et al. developed a multi-objective scheduling model and a hierarchical learning large-neighborhood search algorithm, enabling joint scheduling under uncertain service times [31]. Jo and Son further proposed a path-planning and coordination algorithm for multiple unmanned ground vehicles with heterogeneous tasks in smart farms, improving productivity, efficiency, and robustness through task-priority scheduling [32].

The virtual agricultural AI agent forms a complete functional chain through data aggregation, decision reasoning, information services, and collaborative scheduling,

constituting the "brain" and "nerve center" of the agricultural AI agent system. Although they do not directly touch the soil and crops, they drive the efficient operation of the "perception–decision–execution" closed loop, providing core cognitive support for the intelligent transformation of agriculture. The reviewed studies indicate that the functions of virtual AI agents have extended from simple information processing to the fine organization and flexible arrangement of physical execution ends, achieving a deep transformation from "static decision schemes" to "dynamic operation flows".

2.1.2. Embodied Agricultural AI Agents

Unlike virtual agricultural AI agents, which mainly focus on information processing and cognitive decision-making, embodied agricultural AI agents emphasize autonomous perception and action in the physical world. They are intelligent systems with physical carriers that perceive environmental states through multimodal sensors and complete operational tasks through actuators, thereby forming a closed loop from perception and planning to execution. Due to the complexity of real agricultural scenarios, embodied AI agents face challenges such as energy constraints, environmental adaptation, real-time control, and safety reliability and often rely on edge intelligence to maintain autonomy under communication constraints. Therefore, embodied agricultural AI agents are the core carriers that connect intelligent decision-making with physical execution in agricultural AI agent systems [3,33,34].

1. Direct interaction with the agricultural field environment and execution of physical operations.
   The core value of embodied AI agents lies in their physical execution capability, which enables them to directly interact with agricultural field environments and perform physical tasks beyond the scope of virtual AI agents [35,36]. Jia et al. conducted a systematic review of AI-driven tractor–implement collaborative control and proposed a collaborative control framework integrating agronomic constraints and mechanical dynamics, providing theoretical support for autonomous farm systems [37]. Salah et al. proposed a hybrid-control multi-agent collaborative system that combines pickers' manual skills with robotic precision and navigation capabilities, optimizing the time and cost of apple harvesting through automated fruit-box transportation [38]. Ma et al.'s LSTM-PPO-Weeding framework further integrates mobility and operational control through memory-enhanced reinforcement learning, reducing the average weeding time by up to 74% in greenhouse experiments [39].
2. Replacing manual labor for high-intensity, repetitive, or hazardous field operations.
   The shortage of agricultural labor is becoming increasingly severe, and embodied AI agents have become an important technical path for replacing manual labor in high-intensity, repetitive, and time-sensitive field operations [9,34]. Jo and Son proposed the CBS-HT path-planning algorithm, which assigns priorities according to machine type and operation area. In orchard tests, this method reduced the driving distance of high-priority robots by 13.5% and task time by 13.2%, thereby improving the efficiency of multi-agent operations [40]. Manasherov and Degani proposed an autonomous, multi-agent drone pollination system that jointly optimizes task time and flight distance through target allocation, inventory replenishment, and safe trajectory planning. Their experiments in peach and pear orchards, as well as real drone tests, verified the feasibility of replacing manual pollination [41]. Moreover, Li et al. applied multi-agent reinforcement learning to task planning for multi-arm picking robots, effectively alleviating labor shortages in fruit-picking scenarios [42].

3. Variable operations and precise execution in precision agriculture.
Variable operations are a core requirement of precision agriculture and an important goal of autonomous agricultural machinery. They aim to dynamically adjust operation parameters, such as fertilizer amount, pesticide dosage, and sowing density, according to the spatial variability of soil, crops, pests, and diseases, thereby enabling site-specific and demand-based management [14,36,43]. Embodied AI agents serve as the execution carriers for this requirement. Goral et al. proposed a multi-agent visual system for automatic orchard spraying based on Xception and NasNetLarge, achieving recognition accuracies of 96.88–100% on real orchard image datasets and providing perceptual support for precise spraying [44]. Ankit et al. developed a centralized, multi-agent collaboration framework that optimizes heterogeneous deployment of drones and ground robots through heuristic decision-making, enhancing collaborative operation in automated agricultural scenarios [45]. Li et al. designed a multi-region path-planning framework for precise fertilizer application by improving the DDQN algorithm, demonstrating the potential of deep reinforcement learning in variable-rate operations [46].
4. Collaborating with virtual agents.
Embodied AI agents do not operate independently but collaborate with virtual agents to form a complete intelligent agricultural system. Murad et al. proposed a four-agent collaboration framework in which the perception agent collects environmental data, the decision-making agent generates operation strategies, and the embodied execution device completes task implementation. This framework reflects a "perception–decision–execution" closed-loop collaboration mechanism [10]. Chen et al.'s "Fu Xi Brain" system further extends this collaboration toward full autonomy by integrating a "sky–ground–human–machine" multi-source data collection system with an intelligent decision-making system. In corn planting experiments at Dahewan Farm in Inner Mongolia, this framework achieved an autonomous decision-making accuracy of 89.7% throughout the production cycle, demonstrating the potential of deep collaboration between virtual and embodied AI agents in agricultural autonomous operations [11].

Overall, virtual agricultural agents and embodied agricultural agents differ significantly in their operating mechanisms, advantages, limitations, and roles in agricultural systems. Their comparison is summarized in Table 1.

Table 1. Comparison between virtual agricultural agents and embodied agricultural agents.

Agent Type	Core Mechanism	Advantages	Limitations	System Role
Virtual Agricultural Agents	Data processing, knowledge reasoning, multi-agent collaboration	Strong adaptability, outstanding decision-support capability, high collaborative scalability	Lack of physical execution capability and dependence on data quality	Consultation service agents, planning agents, collaborative scheduling hubs
Embodied Agricultural Agents	Coupling of perception, physical execution	Direct interaction with field environments, autonomous operation capability	Constrained by hardware and environmental conditions	Agricultural robots, precision operation executors, task execution entities

2.2. System Architecture of Agricultural AI Agents

The agricultural AI-agent system is a multi-layer framework that integrates physical resources, intelligent decision-making, collaborative communication, and application services. Drawing on distributed intelligent system architectures and the characteristics of agricultural production environments, this section proposes a four-layer architecture consisting of the infrastructure layer, agent management layer, agent collaboration layer, and application layer. This architecture forms a complete technical chain from resource support and individual intelligence to group collaboration and application services. While the classification system answers "what types of agricultural AI agents exist", the architecture further explains "how these agents are deployed, connected, and collaboratively operated". The overall system architecture of agricultural AI agents is illustrated in Figure 2. Furthermore, recent agricultural AI agent frameworks have explored different approaches to autonomous decision-making, task planning, and multi-agent coordination. Representative frameworks include AgriAgent and Fuxi Brain. AgriAgent emphasizes contract-driven planning and capability-aware tool orchestration, whereas Fuxi Brain focuses on autonomous agricultural decision-making and intelligent task coordination. To further clarify the position of the proposed architecture in the current research landscape, Table 2 compares several representative agricultural AI agent frameworks with the proposed four-layer architecture in terms of architectural focus, planning strategy, collaboration mechanism, and explainability.

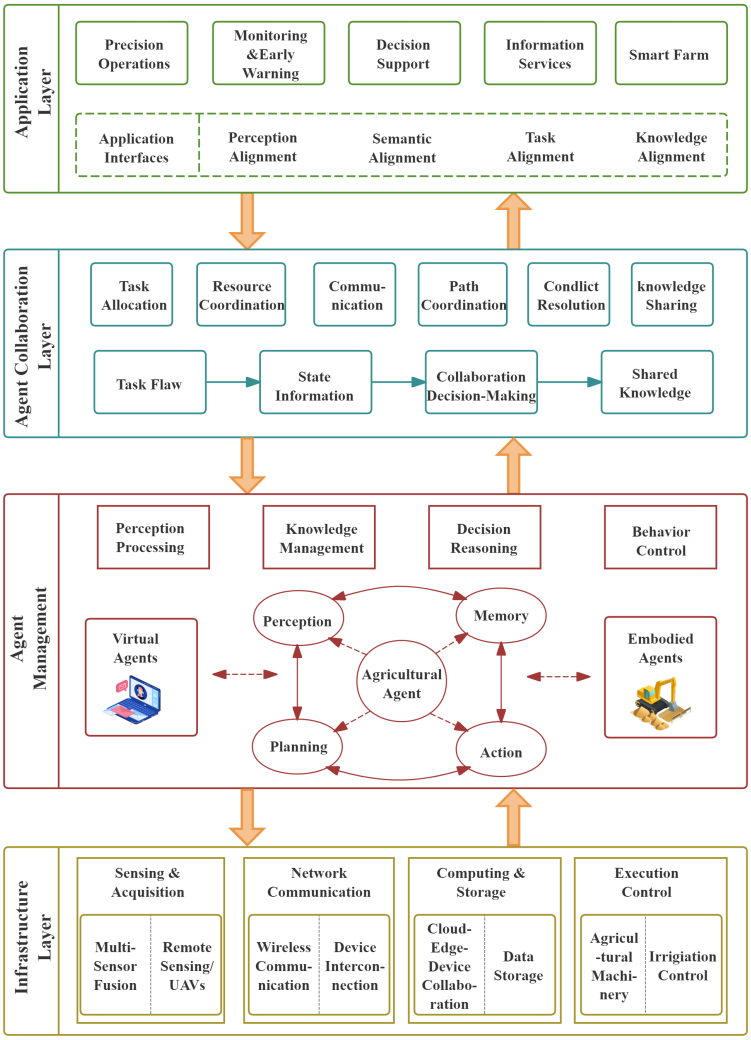


Figure 2. Agricultural AI agent architecture.

Table 2. Comparison of representative agricultural AI agent frameworks.

Framework	Focus	Planning	Collaboration	Explainability
AgriAgent	Real-world agricultural agents	Contract-driven planning	Tool orchestration	High: task contracts
Fuxi Brain	Autonomous decision-making	Intelligent scheduling	Task coordination	Medium: decision support
Proposed Architecture	Four-layer system architecture	Workflow-based planning	Dedicated collaboration layer	High: traceable workflows

2.2.1. Infrastructure Layer

The infrastructure layer is at the bottom of the four-layer architecture and serves as the physical foundation and resource pool of the entire agricultural AI agent system. This layer provides computing resources, communication resources, and sensing and execution resources to the upper layers, shielding the heterogeneity of the underlying hardware and laying a material foundation for the operation and collaboration of the agents. Its core functions include sensing acquisition, network communication, computing storage, and execution control.

In terms of sensing acquisition, the Internet of Things (IoT) and sensor technologies provide the basic data entry point for smart agriculture. Morchid et al. reviewed the application of IoT and sensor technologies in agriculture and proposed a four-layer IoT architecture consisting of the sensing-execution layer, network layer, cloud layer, and application layer. They summarized IoT applications in irrigation monitoring, fertilization management, disease detection, yield and quality monitoring, climate monitoring, and fire detection, covering sensors for soil nutrients, moisture, nitrate, pH, conductivity, CO_2, temperature, humidity, light, water level, livestock, plant diseases, smoke, and flame [47]. Multi-sensor information fusion further compensates for the limitations of single sensors through multidimensional fusion strategies. Zhang et al. reviewed its application in field environmental sensing, analyzed the characteristics, advantages, and disadvantages of different sensors, and highlighted future directions such as precise sensor perception, multidimensional fusion, and agricultural information processing [16].

In terms of network communication, Zhao et al. combined multi-agent theory with GSM communication networks to establish an agricultural irrigation district information system based on multi-agent and GSM. This framework leverages agent intelligence and multi-agent collaborative communication to support irrigation district management and decision-making [48]. Ikidid et al. proposed an intelligent irrigation system based on IoT and agent technology. This approach collects farm environmental data through sensors, calculates daily evapotranspiration using the Penman model, determines the water demand of each farm, and uses multi-agent systems for node control and irrigation parameter configuration. Designed for farmers' associations in rural autonomous regions, it supports irrigation planning based on available water resources, aiming to save water, ensure fair distribution, and maintain crop yields [49].

In terms of computing architecture, Kalyani and Collier explored the role of multi-agent systems in digital twin implementation and analyzed technical gaps in their integration [50]. They further proposed a smart agriculture architecture integrating cloud computing, fog computing, edge computing, multi-agent systems, and digital twins to address resource allocation, resource scheduling, and task scheduling challenges [51]. Kalyani et al. also discussed the application of digital twins in smart crop cultivation through cloud–fog–edge infrastructure, including irrigation, fertilization, nutrient management, and pest and disease control [52]. Zhang et al. systematically reviewed the

applications, challenges, and future directions of digital twin technology in agriculture, covering crop cultivation, pest and disease control, livestock management, agricultural machinery, resource optimization, and decision support. They also identified key technical challenges, including data collection, integration difficulty, and the standardization of three-dimensional crop models [8].

2.2.2. Intelligent Agent Management Layer

The intelligent agent management layer is located above the infrastructure layer and is the individual intelligence core of the agricultural AI agent system. This layer is responsible for the perception processing, decision-making reasoning, knowledge management, and behavior control of individual intelligent agents, directly reflecting the "intelligence quotient" of each agricultural AI agent. Different from the infrastructure layer, which focuses on "what resources are available", the management layer focuses on "what individual intelligent agents can do and will do". The design of this layer directly determines the upper limit of the capabilities of various agricultural AI agents mentioned earlier and is an important technical foundation for achieving autonomous intelligence and high-level automation.

In terms of perception processing, the perception intelligent agent in the four-agent framework proposed by Murad deployed LSTM, GRU, and 1D-CNN prediction models, achieving an accuracy rate of 93.4–96% in soil attribute prediction [10]. Goral et al. proposed a multi-agent visual system for automatic spraying in orchards, using Xception and NasNetLarge deep convolutional neural networks, to achieve target recognition, tree height estimation, and developmental state classification in the automatic spraying scenario of orchards [44].

In terms of knowledge management, Skobelev et al. constructed an agricultural knowledge base based on semantic networks, achieving unified representation and organization of crop production, agricultural resources, and mechanical equipment knowledge, providing knowledge support for precision agricultural management [53]. Ramanathan et al. constructed an agricultural semantic knowledge graph that further describes environmental information and equipment capabilities and supports multi-agent collaborative planning and complex task execution [54]. Additionally, Anand et al. proposed the SEEDS system, which uses RAG technology combined with domain knowledge graphs to provide farmers with precise pest and disease control recommendations, demonstrating the potential application of knowledge-enhanced intelligent agents in agricultural knowledge services [27].

In decision-making reasoning, Agyeman et al. proposed a learning-based irrigation scheduler that integrates machine learning, model predictive control, and multi-agent principles. It uses LSTM to model soil moisture and achieves water savings of 7–23% and irrigation efficiency improvements of 10–35% [22]. The knowledge graph fertilization recommendation and spatio-temporal reasoning methods achieve precise prediction of nitrogen, phosphorus, and potassium fertilization amounts and decision-making under complex conditions through case reasoning, rule reasoning, and spatio-temporal knowledge expression [23,24].

Building on the capabilities of perception processing, knowledge management, and decision-making reasoning, the management layer of agricultural AI agents also needs to address controllability, explainability, and traceability during complex task execution. From a cognitive perspective, agricultural AI agents can be understood as a dual-system paradigm that combines fast reactive perception with deliberative reasoning and long-horizon planning. Inspired by the dual-system cognitive model proposed by Kahneman [55], agricultural AI agents can be organized into two complementary reasoning modes. System 1 is responsible for fast reactive reasoning based on multimodal perception

information, including crop disease detection, environmental anomaly identification, target recognition, and real-time monitoring. These functions are typically supported by computer vision models, sensor fusion algorithms, and lightweight prediction models, enabling rapid responses to dynamic agricultural environments. In contrast, System 2 performs deliberative reasoning for complex agricultural tasks, including resource allocation, irrigation scheduling, agricultural machinery coordination, and long-horizon decision-making. System 2 usually integrates large language models, knowledge graphs, rule reasoning, and planning algorithms to support multi-step analysis and optimization.

To further improve controllability and explainability during complex task execution, recent studies have introduced contract-driven planning mechanisms. In this paradigm, agricultural tasks are represented as explicit task contracts that specify objectives, constraints, required capabilities, expected outputs, and verification criteria before execution begins [56]. For example, a greenhouse disease control task may predefine disease severity thresholds, pesticide dosage constraints, participating agents, execution procedures, and evaluation indicators. Based on these contracts, agents can perform capability matching, workflow orchestration, task decomposition, and execution verification in a structured manner. The combination of dual-system cognition and contract-driven planning provides a practical foundation for agricultural AI agents to achieve reliable multi-step task execution, explainable decision-making, and multi-agent collaboration. It also strengthens the management layer by transforming isolated perception and reasoning capabilities into traceable, controllable, and verifiable intelligent workflows.

2.2.3. Agent Collaboration Layer

The agent collaboration layer is above the management layer and serves as the group intelligence core of the agricultural agent system. This layer is responsible for task allocation, resource coordination, communication interaction, and conflict resolution among multiple agents, achieving a "1 + 1 > 2" collaborative effect. Different from the management layer's focus on "how a single agent does it", the collaboration layer focuses on "how multiple agents work together". This layer is a key support for achieving task-driven cooperation and highly heterogeneous collaboration.

In terms of task allocation and resource coordination, multi-agent task allocation methods have been applied to agricultural labor management and harvest task management. Through task decomposition, role assignment, and resource optimization, they improve the overall configuration efficiency of human, equipment, and operation tasks in agricultural production [57,58]. Guo et al.'s agricultural supply chain negotiation framework further introduces the negotiation mechanism into agricultural resource coordination, achieving dynamic matching and resource collaboration between supply and demand parties through third-party logistics support [59]. For problems such as uncertain service times and task matching constraints in heterogeneous agricultural robot collaborative scheduling, Guo et al.'s AHLLNS algorithm combines hierarchical reinforcement learning and large neighborhood search strategies to achieve joint optimization scheduling of multi-robot systems [31].

In terms of communication interaction, the multi-agent large language model research system summarizes role collaboration, communication protocols, task decomposition, and interaction mechanisms, providing a general framework reference for the collaboration of agricultural agent systems. Jayarathna and Hettige's AgriCom system builds an agricultural community communication platform with multiple roles, including farmers, buyers, sellers, and instructors, enabling timely interaction of agricultural knowledge and market information [29]. Skobelev et al.'s crop management multi-service platform supports crop variety selection, rotation planning, and agricultural plan generation through vertical and

horizontal bidirectional interaction mechanisms of agents, demonstrating the collaborative planning capability in complex agricultural tasks [30].

In practical agricultural deployments, communication interaction is also closely related to edge computing and connectivity economics. Rural agricultural environments often face limited bandwidth, unstable network latency, and constrained computing and energy resources at terminal devices. Therefore, the collaboration layer needs to determine which tasks should be processed locally, which tasks should be offloaded to edge nodes, and which tasks should be uploaded to cloud platforms [60,61]. From the perspective of connectivity economics, the perception task placement strategy helps balance communication cost, communication latency, resource utilization, and energy efficiency. Time-sensitive tasks, such as anomaly detection, obstacle avoidance, greenhouse control, and local path adjustment, can be processed at the terminal or edge side to reduce response latency. In contrast, computationally intensive or long-term optimization tasks, such as model training, knowledge updating, and global scheduling, can be assigned to cloud platforms. Through adaptive task offloading, data compression, event-triggered communication, and edge–cloud resource scheduling, the collaboration layer can reduce unnecessary data transmission and energy consumption while maintaining acceptable service quality in bandwidth-constrained rural environments [62,63].

In terms of path coordination and collaborative control, Jo and Son's CBS-HT algorithm achieves collision-free collaborative operations of heterogeneous agricultural robots through priority safety interval planning, verified in orchard scenarios with high robustness in path coordination [40]. Jogeshwar et al.'s multi-agent coverage planning framework further supports distributed path coordination and coverage optimization in agricultural monitoring tasks [64]. Furthermore, the semi-centralized multi-agent reinforcement learning method proposed by Kalyani and Collier was applied to the collaborative control of irrigation scheduling, achieving water-saving effects superior to fully decentralized control while balancing the advantages of distributed autonomy and centralized coordination [51]. Overall, task allocation, communication interaction, and collaborative control jointly constitute the key support for the emergence of swarm intelligence in the agricultural agent system.

2.2.4. Application Layer

The application layer is at the top of the four-layer architecture and serves as the service portal for agricultural agent systems for end users. This layer encapsulates infrastructure, individual intelligence, and group collaborative capabilities into application services for agricultural business scenarios, serving as the final link in the transformation of technical value into user value. With the development of digital twins, generative artificial intelligence, and intelligent decision-making technologies, the application layer is evolving from traditional single-point functional applications to a full-process intelligent service platform.

In terms of precise operation applications, the intelligent irrigation system developed by Salazar et al. based on a multi-agent architecture and fuzzy logic can autonomously monitor environmental conditions and make decisions on irrigation start and stop, providing an effective solution for water-saving irrigation [65]. The decision-making method of the intelligent irrigation system has also been systematically sorted out, providing a methodological reference for the design of irrigation decision-making systems [66]. The multi-agent visual system for automatic spraying tasks in orchards proposed by Goral et al. achieved an accuracy rate of 96.88% to 100% on real orchard image data, providing perceptual support for precise spraying [44]. Additionally, the LSTM-PPO-Weeding framework proposed by Ma et al. reduced the average weeding time by up to 74% while maintaining

a high success rate, verifying its application potential in agricultural mobile operation tasks [39].

In terms of monitoring and early warning applications, the Chat Demeter system developed by Zhang et al. uses a CNN-Transformer model to achieve leaf disease recognition and disease classification and combines a natural language interaction interface to return diagnostic results and intervention suggestions. The accuracy rate of the validation set is 99.50% and the AUC is 99.91% [26]. Seralathan and Edward's research indicates that the combination of RGB and spectral imaging, transfer learning, data augmentation, and unmanned aerial vehicle platforms helps improve disease monitoring accuracy and resource utilization efficiency [67]. Moreover, digital twin technology can provide a new systematic path for agricultural monitoring and early warning by real-time mapping of agricultural production status [8].

In terms of decision support applications, Chen et al. constructed the "Fuxi Brain" system, integrating agricultural Internet of Things and generative large models, building a "sky–ground–human–machine" full-element data collection and multi-agent collaborative decision-making architecture and achieving an autonomous decision-making accuracy rate of 89.7% for the entire production cycle on the Dahewan Farm in Inner Mongolia [11]. In addition, the online intelligent platform developed by Bahri et al., which is based on multi-agent processing and fuzzy cognitive maps, can provide decision support such as fertilization, crop growth prediction, and irrigation duration estimation for the internal variability of fields, assisting in precision agriculture management [68].

In the field of information service applications, Owiti and Kipkebut proposed a system based on retrieval-augmented generation agents and quantized fine-tuned language models that provides real-time, low-cost, multilingual agricultural advice to small farmers in regions such as Kenya [28]. Anand et al. developed the SEEDS system, which, through domain knowledge graphs and OpenAI models, combined similarity matching to provide precise pest and disease control recommendations, laying the foundation for the question-and-answer framework in precision agriculture [27]. Additionally, Jayarathna and Hettige developed the AgriCom system, which constructed four types of intelligent agents: farmers, buyers, sellers, and instructors, successfully promoting real-time communication among key personnel in the agricultural community [29].

Overall, the application layer represents the ultimate destination of the transformation of agricultural AI agents from technical capabilities to actual value and shows a trend of evolving from single-task applications to a full-chain intelligent service platform covering "precision operation–monitoring and early warning–intelligent decision-making–information service". The above four-layer architecture clearly presents the organizational structure of the agricultural AI agent system: the infrastructure layer provides physical resources, the management layer endows individual intelligence, the coordination layer realizes efficient collaboration among intelligent agents, and the application layer converts technical capabilities into actual value. However, to fully understand the capabilities of agricultural AI agents, it is necessary to summarize from their characteristic dimensions and explore their specific functions and application capabilities to cope with complex agricultural production tasks.

3. Typical Business Operations and Required Model Capabilities of Agricultural AI Agents

The value of agricultural AI agents ultimately does not lie in the complexity of algorithms or models but in their ability to perform specific agricultural production tasks in complex, dynamic, and uncertain environments. Based on the entire agricultural production chain, this chapter systematically discusses the core capability requirements of

agricultural AI agents in different business scenarios from four dimensions: perception and state understanding, knowledge memory and experience management, reasoning and decision-making, and collaborative execution and resource scheduling. Figure 3 presents the capability framework of agricultural AI agents based on these four aspects, and the following sections discuss this framework in detail. These four dimensions are interrelated and progressive, jointly forming a complete business closed loop of "perception–memory–decision–execution". This loop covers the value creation process from environmental information collection, knowledge accumulation, and migration to autonomous task decomposition and multi-agent collaborative operations. Through the systematic presentation of these four dimensions, this chapter reveals that the key to transforming agricultural AI agents from "available" to "usable" lies not in a single technological breakthrough but in the balanced development and organic integration of capabilities across all links of the business closed loop.

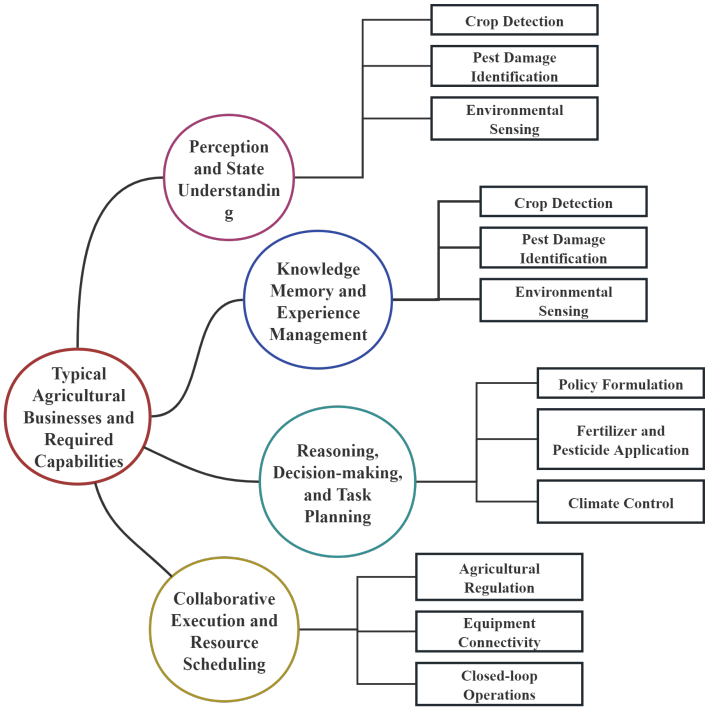


Figure 3. Capability framework of agricultural AI agents

3.1. Perception and State Understanding

Perception and state understanding are the starting point of the agricultural AI agent business chain. This layer extracts environmental states and target attributes from raw physical data through multimodal sensing. In complex agricultural scenarios with changing illumination, cluttered backgrounds, and target occlusion, perception systems need to integrate data alignment, sensor fusion, target detection, segmentation, and state estimation. By forming unified scene representations, agricultural AI agents can identify crops, weeds, pests, growth parameters, stress states, and abnormal events, thereby transforming raw signals into structured knowledge for subsequent knowledge accumulation, task planning, and precise operations.

3.1.1. Crop Detection and Phenotype Analysis

Crop detection and phenotype analysis are the basic business of the perception layer, aiming to obtain key information such as the type, location, and growth status of crops from

image or point cloud data, providing data support for subsequent pest control operations, precise fertilization, and yield prediction.

In crop detection and phenotype analysis, existing research mainly focuses on multimodal perception, cross-scenario generalization, and stress state monitoring. In intensive production scenarios such as nurseries, higher precision is required for perception. Models based on images can fully utilize color and texture information, suitable for leaf disease detection and pest identification; models based on point clouds are better at expressing spatial structure information, suitable for target location and three-dimensional morphology analysis. Therefore, image and point cloud methods have their own advantages in classification, detection, and segmentation tasks, and the appropriate perception scheme should be selected based on specific operation requirements [69].

In complex farmland environments, a single sensor often has insufficient scene adaptability. Studies on visual sensors, laser radar point cloud preprocessing, and multi-sensor fusion methods have shown that multi-source perception fusion can improve the robustness and practicality of crop detection, especially suitable for open scenes with dynamic lighting, complex backgrounds, and occlusions [43]. At the same time, computer vision technology has been integrated into agricultural digitalization processes such as image acquisition, photogrammetry, image analysis, operation decision-making, and planning, providing systematic technical support for crop detection and phenotype analysis [70].

To address the issue of insufficient cross-scenario generalization ability of crop segmentation models, domain generalization and knowledge distillation have become important improvement directions. The two-stage method based on sparse annotation generation and style transfer expansion can enhance the adaptability of crop and weed segmentation models in different farmland scenarios [71]. The knowledge distillation method transfers the knowledge of multiple source domain models to the student model, enabling it to better adapt to unseen real scenarios, and has been shown to improve generalization from simulation to reality on large-scale synthetic datasets [72].

In addition, crop stress monitoring is developing from single image recognition to multi-source remote sensing and the development of intelligent indices. A dynamic framework for vegetation index construction based on reinforcement learning integrates Sentinel-2 multispectral data and smartphone RGB data, achieving early detection of rice stress and having the potential for lightweight and edge deployment, providing a new technical path for low-cost real-time crop management [67].

3.1.2. Pest and Disease Identification

Pest and disease identification is a key task in agricultural sensing operations, and its outcome directly affects the timeliness and accuracy of prevention and control decisions. It is an important link connecting sensing and decision-making. This identification process usually requires multi-level analysis such as disease type identification, severity assessment, and disease spot localization from crop images.

In terms of sensing technology, hyperspectral imaging, as a rapid and non-destructive detection method, demonstrates significant advantages in the early identification of crop diseases. Relevant reviews have systematically analyzed how pathogen infection affects spectral characteristics and pointed out the important role of vegetation indices and machine learning methods in the early detection and warning of diseases [17].

In terms of specific detection methods, pest and disease identification is evolving from traditional image processing to multimodal deep learning methods. For example, detection methods based on spore diffraction texture features can achieve rapid identification using light field information, achieving high accuracy and significantly improving detection efficiency in rice blight detection [73]. At the same time, deep learning methods combine

RGB images with spectral data and improve the model's generalization ability through transfer learning and data augmentation, showing better performance than traditional methods in tasks such as wheat Fusarium blight detection [74].

In terms of model performance optimization, pre-trained convolutional neural networks are widely used in disease classification and severity estimation, and different network structures show differentiated performance in terms of accuracy and complexity [75]. Moreover, the integration of multimodal knowledge graphs and visual models enables pest and disease identification not only to output classification results but also to generate structured diagnostic reports, reflecting the trend from simple identification to knowledge-enhanced interpretation.

In large-scale application scenarios, multi-agent collaborative monitoring frameworks have gradually become an important development direction. Human–machine collaborative systems, by integrating sensing data and the analytical capabilities of agents, achieve real-time monitoring and spatial visualization of pests and diseases [76], providing deployable technical solutions for early warning in large-scale cultivation.

3.1.3. Environmental Sensing

Environmental sensing is a key capability for agricultural AI agents to extend from "being able to see the crops" to "being able to understand the environment". The multi-source information such as soil, weather, and water quality obtained through environmental sensing provides a foundation for irrigation decisions, fertilizer recommendations, and disaster warnings. This sensing process usually relies on distributed sensor networks to collect environmental parameters and achieve comprehensive representation of environmental states through data fusion and analysis.

In terms of multi-modal sensing and data fusion, related research has effectively improved the three-dimensional semantic understanding ability of field scenes by jointly processing RGB images and LiDAR point clouds and introducing cross-modal alignment and perception loss functions. This type of method reflects the trend of agricultural environmental sensing evolving from a single modality to multimodal fusion.

In terms of refined environmental detection, nanomaterial sensors provide important support for microscopic environmental sensing such as pesticide residues. The application of quantum dot sensors and their mechanisms in fluorescence, electrochemical luminescence, etc. has been systematically summarized [77]. In terms of specific implementation, up-conversion fluorescence sensors can achieve highly sensitive detection of organophosphorus pesticides [78], while the multi-sensing method based on SERS and FRET mechanisms supports simultaneous detection of multiple pesticide residues [79,80]. By further combining statistical modeling methods, rapid quantitative analysis of pesticide residues can be achieved [81]. In recent years, related research has also systematically summarized the development directions of sensor stability and miniaturization [82,83], providing effective technical support for precise environmental monitoring.

In terms of large-scale environmental monitoring, remote sensing technology provides a macroscopic perspective for agricultural environmental perception. By comparing the thermal infrared images of unmanned aerial vehicles with satellite remote sensing data, it is shown that unmanned aerial vehicles have higher accuracy in local areas, while satellite data is suitable for regional-scale monitoring. The combination of the two can achieve the collaborative acquisition of multi-scale environmental information [84].

However, current perception studies still show several methodological limitations. Image-based methods are efficient and low-cost, but they are sensitive to illumination changes, occlusion, and background interference. Point-cloud and hyperspectral methods provide richer structural or spectral information, but they usually require expensive

sensors and higher computational costs. Multi-sensor fusion can improve robustness, yet it also introduces challenges in data alignment, calibration, real-time processing, and deployment maintenance. Therefore, future research should not only improve perception accuracy but also compare different sensing schemes under real agricultural conditions and develop lightweight, robust, and low-cost perception solutions for agricultural AI agents. Overall, crop detection and phenotypic analysis, pest and disease identification, and environmental sensing jointly constitute the perception capability of agricultural AI agents. Current research is moving from single-target recognition toward multimodal fusion, cross-scale monitoring, and semantic understanding. However, perception outputs can support autonomous decision-making only when they are continuously stored, organized, and transformed into reusable knowledge. Therefore, effective knowledge memory and experience management mechanisms are essential for enabling long-term learning and experience-driven agricultural intelligence.

3.2. Knowledge Memory and Experience Management

Knowledge memory and experience management form the supporting layer of the agricultural AI agent business chain. This layer stores, organizes, retrieves, and reuses historical data, domain knowledge, and operational experience. Unlike the perception layer, which processes real-time environmental information, the knowledge memory layer transforms discrete experiences into reusable knowledge assets. It supports cross-season accumulation, cross-scenario transfer, traceable decision evidence, and the transition from reactive execution to experience-driven intelligence.

3.2.1. Land Parcel Archive Management and Knowledge Base Construction

Land parcel archive management and knowledge base construction are the basic tasks for agricultural AI agents to achieve knowledge memory and experience management. Their core objective is to organize and associate spatial information, attribute characteristics, and historical agricultural activities in a structured manner, providing reliable data support for subsequent decision-making reasoning. With the integration of multi-agent technology and knowledge management methods, research in this area has gradually evolved from single data management to multi-scale modeling and knowledge-driven management.

In farm modeling and policy evaluation, multi-agent methods are used to depict the behavior of farm entities and their interaction relationships. Oudendag et al. modeled each farm as an independent agent to simulate the impact of agricultural policy changes on farm behavior, demonstrating the application value of agent modeling in policy evaluation [85]. The method utilized multi-agent simulation to analyze the impact of land transfer among farmers on agricultural sustainability, revealing the relationship between transaction costs, land leasing markets, and land abandonment [86].

In spatial multi-agent modeling, research further focuses on spatial heterogeneity and social interaction factors. Berger proposed a spatial multi-agent model to simulate farmer interactions, innovation diffusion, and changes in resource utilization [87]. In land use optimization, Kalyani modeled different crop types as intelligent agents and optimized the spatial layout of crops under agronomic, economic, and water resource constraints, achieving a balance between resource utilization efficiency and ecological risk [51].

In knowledge base construction, agricultural knowledge management gradually shifts from data storage to knowledge organization and knowledge discovery. Imane combined the multi-agent framework with variable clustering and association rule mining to discover key factors affecting the cost of olive oil, demonstrating the potential of multi-agent data mining in agricultural knowledge discovery [88]. Skobelev constructed an agricultural knowledge base based on ontologies and semantic networks, organizing knowledge related

to crop production, agricultural resources, and mechanical equipment, supporting the storage, editing, verification, and visualization of knowledge [53]. In recent years, digital twin technology has further provided a new paradigm of virtual–real integration for land parcel archive management, mapping the physical attributes and historical data of land parcels to virtual space to achieve dynamic updates and continuous accumulation of knowledge assets [8].

However, existing studies on land parcel archives and knowledge base construction still have limitations. Multi-agent simulations can represent farmer behavior and land-use interactions, but they often rely on simplified assumptions and lack validation in real production environments. Ontology-based knowledge bases improve knowledge organization, yet they depend heavily on expert-defined rules and may adapt poorly to dynamic field changes. Digital twins support continuous record updating but require stable data streams, standardized interfaces, and high-quality historical data. Therefore, future research should integrate land parcel archives, knowledge bases, and digital twins into an adaptive memory system for agricultural AI agents, rather than treating them as isolated data management tools.

3.2.2. Knowledge Retrieval and Intelligent Question Answering

With the continuous growth of agricultural data volumes, how to efficiently retrieve precise agricultural knowledge from massive heterogeneous information has become an important challenge for intelligent agricultural systems. Knowledge retrieval and intelligent question answering, as important applications of the knowledge memory capability of agricultural AI agents, integrate natural language processing and information retrieval technologies to provide personalized and understandable decision support services for farmers and related practitioners, thereby enhancing the intelligence level of agricultural management.

In terms of information retrieval frameworks, personalized retrieval methods based on semantic web and agent technology provide new ideas for agricultural information acquisition [89]. Such methods introduce semantic annotations to web content and combine agents to model user needs and filter results, effectively improving the information acquisition efficiency of multi-role users in the agricultural field.

At the specific application level, agent-driven recommendation systems are used in agricultural knowledge services. For example, the multi-scale pest recommendation system integrates growth models, invasion models, and information scale conversion mechanisms to simulate pest development trends and assist experts in evaluating and generating control strategies [90]. This method reflects the transition from “information retrieval” to “knowledge recommendation and decision support”.

In terms of technological development trends, conversational artificial intelligence and retrieval-enhanced generation (RAG) technologies are gradually becoming important supports for agricultural intelligent question answering. Relevant studies show that vector databases and RAG frameworks can play a key role in storing conversation context, integrating domain knowledge, and improving response accuracy [91], providing a technical foundation for agricultural AI agents to achieve long-term memory and context perception.

3.2.3. Historical Tracking and Traceability

Historical tracking and traceability are important components in precision agriculture, focusing on the management of the entire agricultural production process in a recordable, queryable, and verifiable manner, thereby providing support for agricultural product quality safety, resource utilization optimization, and decision evaluation. By recording the operation trajectories, environmental parameters, and decision-making processes during

the crop's entire life cycle, agricultural AI agents can achieve closed-loop management from "process visibility" to "result traceability", providing a basis for post-event analysis and responsibility tracing.

At the implementation level, multi-agent technology has been applied to the agricultural quality safety traceability system. Through the division of labor and collaboration of different functional agents, it realizes quality warning and supervision control of the agricultural supply chain, thereby enhancing the response ability and reliability of the framework [92]. Further, the multi-agent modeling method combining information entropy and complexity theory is used to analyze the strategic game and evolution process of multiple agents in the agricultural quality safety system, revealing the key role of regulatory mechanisms and intermediary organizations in system stability [93].

In terms of cross-regional traceability and resource scheduling, blockchain technology provides important support for the trusted storage and sharing of agricultural data. For example, storing agricultural machinery resource information in a distributed file system and recording scheduling results through blockchain can achieve data immutability and full-process traceability, thereby enhancing the transparency and credibility of agricultural resource scheduling [94]. Additionally, the multi-agent collaborative model based on evolutionary game theory proposes an "incentive–constraint–information" collaborative optimization path from the perspective of mechanism design, providing theoretical support for the transformation from passive response to active collaboration in agricultural disaster response [95].

Overall, agricultural knowledge memory and experience management are shifting from static data storage to dynamic knowledge evolution and intelligent services. Land parcel archives, knowledge bases, retrieval systems, and traceability mechanisms jointly support data accumulation, knowledge reuse, process verification, and responsibility attribution. However, current studies still pay insufficient attention to the continuous validation and updating of historical experience, the reliability of retrieval results, and the connection between traceability records and autonomous reasoning. Therefore, future research should integrate archives, knowledge graphs, RAG-based retrieval, and traceability records into a unified long-term memory mechanism for agricultural AI agents, supporting reliable reasoning, adaptive decision-making, and continuous optimization in real agricultural environments.

3.3. Reasoning Decisions and Task Planning

Reasoning, decision-making, and task planning form the core of the agricultural AI agent business chain. This process enables agents to analyze sensed information, historical data, and knowledge-based experience and then transform them into executable operation plans. It supports both single-task decisions, such as irrigation, fertilization, and pesticide application, and complex planning tasks, such as crop rotation, planting density optimization, and resource allocation. By adjusting decisions according to dynamic farmland conditions, agricultural AI agents can improve operational efficiency, resource utilization, and production sustainability.

3.3.1. Irrigation Decision

Irrigation decision is a fundamental task in the reasoning decisions of agricultural AI agents, and the quality of the decision directly affects the efficiency of water resource utilization and crop yield. An irrigation decision task is usually based on soil moisture conditions, crop water requirement characteristics, and meteorological conditions, optimizing the timing and amount of irrigation.

From the perspective of the method system, agent technology provides complete support from sensing to decision-making for irrigation decisions. Relevant reviews indicate that intelligent agents with planning, learning, and collaborative capabilities can achieve adaptive irrigation scheduling based on the spatiotemporal changes of the soil–plant–atmosphere system, thereby significantly improving water resource utilization efficiency [96]. At the same time, irrigation decision methods have evolved from a single model to a multi-stage integrated framework, covering key links such as soil moisture monitoring, need model construction, and control strategy optimization, providing a methodological basis for systematic design [66].

At the application level, Internet of Things and big data technologies have promoted the practical application of intelligent irrigation systems. For example, by extracting soil image features using convolutional neural networks and combining fuzzy control and expert rule libraries, irrigation and fertilization can be optimized in coordination [97]. Further, a learning-based irrigation scheduler that integrates machine learning and model predictive control, using LSTM to model soil moisture dynamics and combining a multi-agent MPC framework, has achieved a 7–23% water-saving effect and a 10–35% efficiency improvement in actual farmland [22].

In terms of algorithm development trends, deep reinforcement learning has gradually become an important technical path for intelligent irrigation, enabling adaptive optimization of strategies in dynamic environments and improving decision-making capability under uncertain conditions [9]. Additionally, in water resource management in arid areas, the modeling method based on intelligent agents is used to evaluate the transformation effect of irrigation systems, achieving significant improvements in efficiency [98]. Research combining multi-agent optimization and game models further indicates that by introducing comprehensive indicators such as water–carbon footprint, it is possible to achieve coordinated optimization of water saving, emission reduction, and yield increase [99].

However, existing irrigation decision studies still show several limitations. Rule-based and fuzzy-control methods are interpretable and easy to deploy, but they often have limited adaptability under changing weather and soil conditions. Machine learning and model predictive control can improve dynamic scheduling, yet they require reliable historical data and accurate system modeling. Deep reinforcement learning provides stronger adaptive optimization ability, but its training process is data-intensive and its decisions are sometimes difficult to interpret. Therefore, future irrigation decision research should further balance water-saving performance, model interpretability, deployment cost, and robustness under real field conditions.

3.3.2. Fertilizer and Pesticide Application Decision-Making

Fertilizer and pesticide application decision-making is the core application scenario of agricultural AI agents in precision agriculture. The results of decision-making tasks directly affect the quality of agricultural products and the sustainability of the ecological environment. Such tasks usually comprehensively optimize the timing, dosage, and operation path of fertilizer and pesticide application based on soil nutrient conditions, crop nutrient requirements, and the severity of pest and disease occurrences.

From the perspective of optimization modeling, the fertilizer and pesticide application decision-making problem often has multi-objective and multi-constraint characteristics, involving the trade-off between operation efficiency, resource consumption, and environmental impact. Related research usually models the precision agriculture system as a multi-agent system, achieving complex task decomposition and execution through the collaboration of different functional entities. On this basis, a hybrid optimization method combining genetic algorithms and particle swarm optimization is used for multi-objective

decision-making, achieving balanced optimization of efficiency and sustainability in fertilizer and pesticide application operations [100].

In terms of decision-making methods, agricultural AI agents are evolving from traditional rule-driven to a combination of data-driven, knowledge-driven, and model-driven approaches. With the development of multi-modal perception and intelligent decision-making technologies, the joint modeling of visual, language, and environmental data provides more comprehensive information support for fertilizer and pesticide application. Existing research uses knowledge graphs and case reasoning to implement fertilizer application plan recommendations, enhancing the interpretability of agricultural decisions [23]; at the same time, the development of agricultural large language models and embodied intelligence further promotes the closed-loop connection between decision-making suggestions and actual operation execution [25].

In terms of system development trends, fertilizer and pesticide application decision-making is gradually integrating with irrigation scheduling, forming a water–fertilizer integrated collaborative decision-making framework. Intelligent decision-making methods based on the spatiotemporal changes of the soil–plant–atmosphere system provide important references for the coordinated optimization of water and fertilizer resources [96]. These developments reflect the trend of agricultural decision-making evolving from single-task optimization to multi-task collaboration and system-level optimization.

3.3.3. Greenhouse Regulation

Greenhouse regulation is a typical comprehensive decision-making scenario for agricultural AI agents in facility agriculture, involving complex problems such as coupling of multiple environmental variables, conflicts of multiple objectives, and real-time responses. Greenhouse regulation typically involves dynamic adjustment of key factors such as temperature, humidity, CO_2 concentration, and light to optimize resource utilization efficiency while meeting the growth requirements of crops.

From the perspective of technological development, greenhouse regulation methods are evolving from traditional control strategies to intelligent and systematic directions. Relevant reviews indicate that model predictive control (MPC), multi-agent reinforcement learning, and digital twin technologies have gradually become important technical paths for greenhouse intelligent regulation, providing a unified framework for energy optimization and system modeling [101]. At the same time, the development of greenhouse control strategies has gone through an evolution process from classical PID control to model-driven control and then to data-driven methods, providing theoretical support for precise regulation in multi-variable coupled environments [25].

At the platform implementation level, reinforcement learning methods are widely applied to greenhouse climate regulation problems. For example, the control model based on the TD3 algorithm effectively improves the robustness of the system in complex climate conditions and achieves stable Pareto optimization between yield and resource consumption through the introduction of feature selection mechanisms and re-training strategies for environmental disturbances [102]. This method reflects the transition from static control to adaptive optimization control.

Overall, irrigation, fertilization and pesticide application, and greenhouse regulation represent three typical decision-making scenarios for water management, input configuration, and facility environment control. Existing studies show that agricultural decision-making methods are evolving from rule-based and single-variable optimization toward multi-source information fusion, multi-objective trade-offs, and closed-loop adaptive regulation. However, several research gaps remain. First, many methods achieve good performance in specific scenarios but lack cross-region, cross-crop, and long-term vali-

dation. Second, optimization models often emphasize efficiency or yield improvement, while insufficiently considering interpretability, uncertainty, operational risk, and farmer acceptance. Third, decision outputs are still weakly connected with executable machinery control, feedback validation, and continuous knowledge updating. Therefore, future research should strengthen the integration of reasoning models, agronomic knowledge, operational constraints, and execution feedback, so that agricultural AI agents can move from generating decision suggestions to supporting reliable and adaptive task planning in real agricultural environments.

3.4. Collaborative Execution and Resource Scheduling

Collaborative execution and resource scheduling form the implementation layer of the agricultural AI agent business chain, converting decision plans into physical operations. This layer focuses on task allocation, resource scheduling, path coordination, and conflict resolution among multiple agents and heterogeneous equipment. Unlike the decision-making layer, which determines "what to do and when to do it", the execution layer addresses "who does it, how it is done, and how agents collaborate efficiently". It is essential for enabling large-scale, automated, and reliable agricultural operations in dynamic and unstructured field environments.

3.4.1. Tractor Scheduling and Path Planning

Tractor scheduling and path planning are the basic business of the collaborative execution layer. Their goal is to convert operation plans for irrigation, fertilization, pesticide application, and related tasks generated by the decision-making layer into executable tasks for agricultural machinery, robots, and unmanned aircraft, and optimize task allocation, operation sequence, and driving paths under the requirements of operation demands, equipment status, and resource constraints.

From the perspective of the macro technical framework, the precise crop management under the background of Agriculture 5.0 is integrating emerging technologies such as collaborative robots, 6G, digital twins, Internet of Things, blockchain, cloud computing, etc., providing a systematic development direction for tractor scheduling and path planning [103]. In terms of cross-regional tractor resource scheduling, blockchain, smart contracts, and distributed storage technologies can be used to achieve trusted scheduling of tractor resources and full-process traceability, thereby improving resource utilization and reducing cross-regional operation costs [94].

In terms of multi-robot collaborative path planning, deep reinforcement learning and intelligent optimization algorithms have become important technical paths. To address issues such as overestimation of Q-values, redundant samples, and insufficient exploration efficiency in multi-machine agricultural scheduling, the improved DQN method can reduce energy consumption, system delay, and operation time in simulation scenarios such as cotton harvesting [104]. In heterogeneous robot scheduling, heuristic optimization, mixed integer linear programming, and priority path planning methods are used to solve problems such as task allocation, workload balancing, human–machine collaboration, and traffic conflicts [105–107]. At the same time, research on tractor obstacle avoidance and autonomous navigation technology provides support for safe operations in complex field environments [33,35].

In terms of unmanned aircraft cluster scheduling, multi-agent reinforcement learning and attention mechanisms are used to enhance cluster collaboration capabilities. Enhanced multi-agent cluster control algorithms can achieve dynamic adjustment of patrol paths, real-time obstacle avoidance, and path optimization through virtual navigators, enhancing the stability of unmanned aircraft clusters in complex environments such as orchards [108].

The multi-agent soft actor-critic algorithm based on Transformer models uses self-attention mechanisms to model the relationships among unmanned aircraft, improving the convergence speed and completion efficiency of collaborative data collection tasks [109]. In the field of ground unmanned vehicle navigation, multi-sensor fusion and graph neural networks provide new ideas for high-precision positioning and cluster collaboration. The mapping and positioning method that integrates LiDAR, GNSS, and inertial measurement units can alleviate the problem of insufficient satellite signals in farm environments, fields, and roads, achieving centimeter-level positioning accuracy [110]. The self-supervised graph neural network model can be used for cluster robot navigation and landing prediction, demonstrating the application potential of graph structure learning in multi-robot collaborative navigation [111].

3.4.2. Equipment Interconnection and Heterogeneous Collaboration

Equipment interconnection and heterogeneous collaboration are important extensions of the collaborative execution layer, aiming to achieve interconnection and task collaboration between different types and functions of equipment. In agricultural scenarios, equipment heterogeneity mainly manifests in the collaborative operations of aerial drones and ground robots, differences in communication protocols of different brand agricultural machinery, and the collaborative requirements between perception devices and execution devices. The core of such tasks lies in achieving efficient collaboration among multiple devices through unified scheduling and collaborative control mechanisms, thereby enhancing the overall operational efficiency and robustness of agricultural operations.

From the perspective of equipment collaboration and perception execution, agricultural drones, as typical embodied devices, have become the focus of research due to their autonomous obstacle avoidance and collaborative operation capabilities. Relevant reviews indicate that in the Internet of Things environment, agricultural spraying drones still face challenges in data collection, real-time processing, and obstacle avoidance under dynamic operating conditions, requiring a balance between perception accuracy and response speed [112].

In the field of agricultural supply chain collaboration, multi-agent methods are used to model the collaborative relationships among different entities. Relevant research has achieved the optimization of agricultural product logistics resources and collaborative value-added through the construction of a multi-agent collaborative management framework [113]; at the same time, the multi-agent electronic contract mechanism, based on multi-agent technology, supports dynamic collaboration between producers and processors through methods such as combined auctions, demonstrating the application potential of agent technology in the agricultural supply chain [114].

At the resource management and system collaboration level, multi-agent reinforcement learning and game theory methods are used to solve the problems of task offloading and resource allocation among heterogeneous devices. The scheduling strategy based on the random game framework can achieve collaborative optimization between quality of service and energy consumption and shows good convergence and stability in multi-scale robot systems [61]. Moreover, the integration of multi-agent systems and digital twin technology provides a virtual–real combined system support for equipment interconnection, making the collaborative scheduling and state prediction in complex agricultural scenarios more controllable and interpretable [50].

In terms of collaborative control methods, distributed control and group collaboration mechanisms have become key research directions. For nonlinear multi-agent systems, the adaptive leader–follower consensus control method can achieve stable collaboration among multiple robots without centralized control [115], while the hierarchical collaborative

control strategy uses topological structure modeling to support group collaboration in multi-crop and multi-task scenarios, providing a theoretical basis for heterogeneous device collaboration in complex agricultural environments [116].

However, existing studies on equipment interconnection and heterogeneous collaboration still face several practical limitations. Drone–ground robot collaboration can improve operational coverage and flexibility, but it depends heavily on stable communication, accurate localization, and real-time task coordination. Blockchain and smart-contract-based scheduling improve trust and traceability, yet their deployment may introduce additional computational and maintenance costs. Distributed control and multi-agent reinforcement learning enhance autonomy, but their robustness under unstable field networks, device failures, and heterogeneous communication protocols remains insufficiently validated. Therefore, future research should pay more attention to interoperability standards, fault tolerance, communication reliability, and low-cost deployment when designing collaborative execution systems for agricultural AI agents.

3.4.3. Closed-Loop Operations and Collaborative Optimization

Closed-loop operations and collaborative optimization are the deepening business of the collaborative execution layer. Their core lies in the organic integration of perception, decision-making, execution, feedback, and re-perception, forming a continuous optimization mechanism to achieve dynamic adjustment and long-term performance improvement of agricultural operations.

From the perspective of system architecture, the self-organizing decision support system based on multi-agent architecture provides an important implementation path for closed-loop optimization. Relevant research combines the Internet of Things and edge computing, and, through methods such as swarm intelligence, cellular computing, genetic algorithms, and Pareto optimization, this approach realizes adaptive regulation and multi-objective optimization of the dynamic agricultural environment [117]. This framework continuously refines decision strategies through real-time data feedback, reflecting the transformation of agricultural systems from static execution to dynamic optimization.

At the specific application level, multi-vehicle collaborative operations provide a typical scenario for closed-loop optimization. Relevant research introduces task priority mechanisms and multi-index evaluation systems to achieve collaborative path planning and dynamic adjustment of multiple unmanned ground vehicles in complex farmland environments, thereby improving overall operational efficiency and reducing operation conflicts [32]. Further, from the perspective of system-level collaboration, the multi-dimensional collaborative framework unifies modeling of data, processes, and organizational levels to achieve full-chain optimization and efficient resource allocation of agricultural operations [118], providing technical support for the development of agricultural production models toward more refined and intensive management.

Overall, collaborative execution and resource scheduling are evolving from single-device control toward multi-agent collaboration and system-level optimization. Despite recent progress, challenges remain in real-world robustness, heterogeneous device coordination, and the integration of feedback-driven optimization. Future research should strengthen the coupling among task planning, equipment execution, feedback validation, and resource scheduling to support reliable, scalable, and self-optimizing agricultural AI agent deployment.

3.5. Summary: Transition from Single-Point Technology Breakthrough to Full-Chain Business Closed-Loop

This chapter reviews the capability evolution of agricultural AI agents from four dimensions: perception, knowledge, decision-making, and execution. Together, these

dimensions form a full-chain technical framework from data acquisition to physical operation. The perception layer supports crop and environmental state understanding through computer vision, hyperspectral imaging, and multi-sensor fusion. The knowledge layer builds structured and traceable knowledge systems based on knowledge graphs, blockchain, and multi-agent simulation. The decision-making layer enables intelligent planning through deep reinforcement learning, model predictive control, and large model inference. The execution layer promotes efficient collaboration and dynamic adjustment of heterogeneous equipment through path planning, multi-machine scheduling, and closed-loop optimization.

To further summarize the relationship among these capability modules, Figure 4 illustrates the evolution of agricultural AI agents from single-point intelligence to full-chain closed-loop intelligence. As shown in the figure, perception, knowledge memory, reasoning decision-making, collaborative execution, and feedback self-evolution are not isolated technical modules but are connected through continuous data flow and feedback mechanisms, forming a complete closed-loop capability chain.

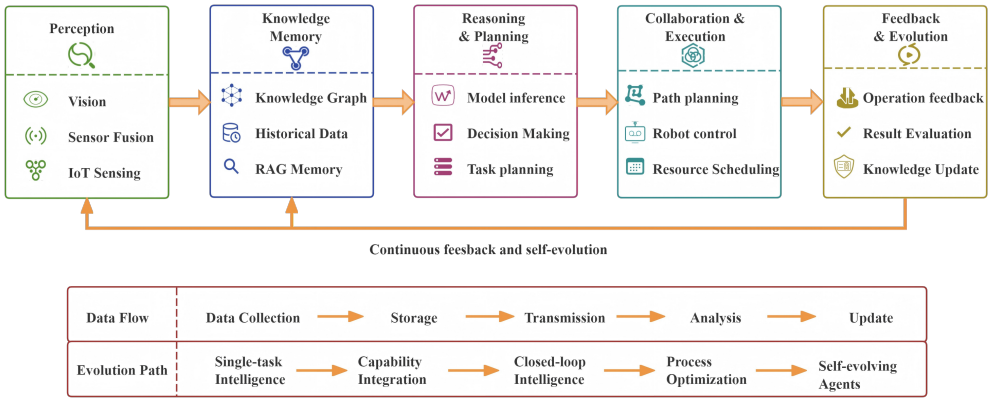


Figure 4. Evolution from single-point to closed-loop agricultural AI agents.

Overall, agricultural AI agents are evolving from "single-point intelligence" to "system intelligence", with their capability boundaries expanding from single-task optimization to multi-task collaboration and full-process closed-loop optimization. The collaborative integration of various technologies not only significantly improves agricultural production efficiency and resource utilization levels but also provides key support for achieving highly reliable, adaptive, and sustainable intelligent agricultural systems and lays the foundation for the subsequent development of agricultural AI agents toward self-learning and self-evolution.

4. Workflow Examples of Agricultural AI Agents: Full-Process Task Organization Based on Model Capabilities Required for Typical Tasks

An agricultural AI agent workflow is not simply a collection of AI models or a fixed process for a specific crop or device. Rather, it organizes capabilities such as perception, diagnosis, decision-making, planning, execution, and feedback into reusable task chains through task orchestration. In essence, it is a capability organization mechanism that transforms agricultural tasks, including pest control, water–fertilizer management, greenhouse regulation, machinery operation, and agricultural consulting, into continuous processes that can be understood, decomposed, matched, executed, and updated by the agent system.

The previous sections explained the existing forms, business requirements, and organizational hierarchy of agricultural AI agents from the perspectives of classification systems, system architecture, and model capabilities. Based on this foundation, this sec-

tion further answers the question of "how an agricultural AI agent completes a specific agricultural task," constructing an agricultural AI agent workflow of "business requirement identification—model capability matching—agent selection—task decomposition and orchestration—collaborative execution—result feedback and knowledge update," starting from the capabilities required for typical tasks. To facilitate subsequent illustration, this section also provides a logical framework suitable for translation into a diagram. As shown in Figure 5, the left side represents typical agricultural tasks, the middle represents required model capabilities, and the right side represents the agent workflow links, ultimately forming a closed-loop system oriented toward overall agricultural tasks.

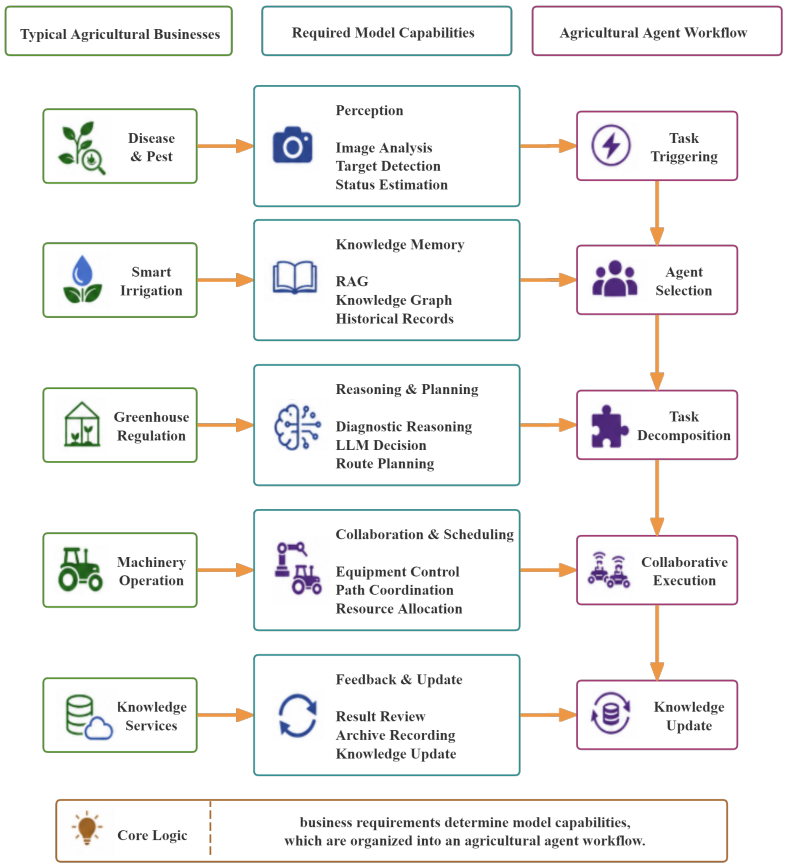


Figure 5. Agricultural business–capability–workflow mapping framework.

4.1. Model Capability Requirements for Typical Agricultural Tasks

Agricultural production is characterized by complex scenes, continuous tasks, diverse targets, and dynamically changing environments. For an agricultural AI agent system to truly support production tasks, it must first answer the question of "what model capabilities are required for the target agricultural task." For example, pest and disease control tasks require image recognition models to detect abnormal leaves, fruit lesions, or pests; diagnostic models to judge disease types and severity; knowledge retrieval or knowledge graph models to provide prevention and control basis; and decision-making models to generate treatment plans involving pesticides, dosages, ventilation, and isolation. Similarly, smart irrigation requires sensor data fusion, evapotranspiration estimation, soil moisture prediction, irrigation decision-making, and valve control capabilities; greenhouse environment regulation requires environmental state perception, crop growth modeling, climate control, and equipment linkage capabilities; agricultural machinery operations require environmental perception, path planning, trajectory tracking, obstacle avoidance, and multi-machine collaborative control capabilities.

Therefore, the workflow design of agricultural AI agents should not start directly from a "model inventory," but from the mapping relationship of "task—capability—agent." A review of agricultural object detection shows that object detection has already served multiple business types, such as crop monitoring, weed management, pest detection, and autonomous field operations, essentially providing visual perception capabilities for subsequent agricultural tasks [119]. A precision weeding review further points out that machine vision and deep learning are driving tasks like weed recognition, variable-rate weeding, and precision plant protection from uniform operations to targeted operations [120]. In terms of disease detection, YOLOv8-GDCI proposed an improved object detection model for different infected parts of pepper blight, illustrating that crop disease recognition capabilities can be adapted for specific agricultural targets [4]. These literatures do not all directly discuss "agents," but they provide the most fundamental model capabilities in the agricultural AI agent workflow.

Although these studies do not all directly take "agricultural AI agents" as their research object, they correspond to key capabilities in the agent workflow, such as perception, diagnosis, decision-making, and execution. Based on the above analysis, this paper further summarizes typical agricultural tasks, core questions, required model capabilities, and corresponding agent types. From a comparative perspective, different agricultural businesses do not require the same type of intelligence. Pest and disease control depends more heavily on visual perception, diagnostic reasoning, and knowledge retrieval; smart irrigation emphasizes time series prediction, water demand estimation, and closed-loop control; greenhouse regulation requires continuous state modeling and multi-equipment coordination; agricultural machinery operations rely on path planning, trajectory tracking, obstacle avoidance, and safety constraints. Therefore, model capability analysis should not be understood as a list of available algorithms but as a task-oriented mapping between agricultural business requirements and agent roles. A major limitation of current studies is that many models are still evaluated as isolated components, while their compatibility with downstream decision-making, equipment execution, and feedback updating is rarely verified in complete agricultural workflows.

4.2. Task Triggering: From Business Needs to Agent Task Generation

In the agricultural AI agent workflow, task triggering is the starting point for business needs to enter the agent system. Task triggering is not equivalent to simple data collection, but transforms abnormal states, management needs, or user intentions in agricultural scenes into executable tasks. According to the source of triggers, they can be divided into four categories: First, image triggering, where visual models recognize abnormal leaf spots, fruit rot, pests, weeds, and obstacles, thereby triggering diagnostic or operational tasks. Second, sensor triggering, where parameters such as temperature, humidity, soil moisture, light, CO_2 concentration, and equipment height deviate from thresholds, triggering regulation tasks. Third, business rule triggering, where preventive tasks are triggered by farming calendars, crop growth stages, weather forecasts, and pest epidemic rules. Fourth, human–machine interaction triggering, where farmers propose consultation or management requests via natural language, which the system parses into Q&A, diagnostic, or scheduling tasks.

Image triggering is particularly important in pest control and precision operations. Chat Demeter completes diseased leaf recognition, instance segmentation, and disease classification through real-time leaf image acquisition and a CNN-Transformer model and returns diagnostic results and treatment recommendations via a natural language interface, embodying a typical flow of "image input—anomaly recognition—task triggering—diagnostic recommendation" [26]. A review of agricultural object detection points out

that object detection has evolved from merely recognizing crops or weeds to a fundamental capability supporting crop monitoring, weed management, pest detection, and autonomous field operations [119]. Therefore, from an agent perspective, the role of an image model is not just to output category labels but to convert visual anomalies into subsequent diagnostic, decision-making, and execution tasks.

Sensor triggering is more suitable for continuous operations such as water and fertilizer management, greenhouse regulation, and equipment control. A review of smart sprinkler irrigation notes that soil moisture sensors, wireless sensor networks, remote data management, and autonomous sprinkler control constitute an important foundation for remote autonomous irrigation systems [121]. A greenhouse intelligence review shows that greenhouse climate control needs to integrate multiple sources of information, including temperature, humidity, light, CO_2 crop growth status, and control strategies [25]. Research on boom height detection further explains that sensory feedback from the execution equipment itself can also become a critical input for smart operations, aimed at improving the reliability and adaptability of spraying tasks [5]. Thus, the essence of the task triggering layer is to transform changes in the agricultural environment and equipment status into operational tasks that agents can process.

4.3. Agent Selection and Capability Matching: From Model Capabilities to Role Division

After a task is triggered, the agricultural AI agent system needs to select the appropriate agents based on the task content. Unlike traditional agricultural information systems, agricultural AI agent systems typically do not rely on a single model to complete all tasks, but achieve complex task processing through role-based agent division. A complete agricultural workflow usually requires perception agents, diagnosis agents, knowledge agents, decision agents, planning agents, execution agents, and supervisory agents. Among them, perception agents handle image, sensor, and remote sensing data; diagnosis agents judge pests, diseases, environmental stress, and equipment anomalies; knowledge agents invoke knowledge graphs, rag, case bases, and agronomic rules; decision agents generate treatment plans; planning agents arrange paths, scheduling, and resources; execution agents invoke equipment; and supervisory agents check task status, coordinate conflicts, and aggregate feedback.

Existing agricultural AI agent research has already reflected this capability-matching ideology. The Agentic AI framework designs soil monitoring, weather perception, visual disease detection, and supervisory chatbots as different agents, achieving information aggregation and predictive analysis through a supervisory agent [10]. The “Fuxi Brain” further integrates agricultural IoT with generative large models to build a “space–air–ground–human–machine” all-factor data collection system and multi-agent collaborative decision-making architecture, reflecting the holistic trend of agricultural production from data collection to autonomous decision-making [11]. Related research illustrates that the key to an agricultural AI agent workflow is not having one model complete all tasks but matching agents according to the required capabilities of the business.

From the research of Jiangsu University and related agricultural engineering fields, many models, although not directly using the term “agent,” can be incorporated into workflows as agent capability modules. For example, object detection reviews and pepper blight recognition models can support perception and diagnosis agents [4,119]; smart irrigation decision reviews and greenhouse intelligence reviews can support decision and regulation agents [25,66]; orchard robot path planning and multi-scenario autonomous navigation reviews can support planning and execution agents [122,123]; intelligent orchard spraying, variable-rate spraying, and agricultural machinery collaborative control research can support execution and collaborative agents [36,37,124]. Thus, single-point technologies

in the agricultural engineering field can be organized into agricultural AI agent workflows via the "model capability—agent role" approach.

4.4. Dynamic Task Orchestration and Matching: The Concatenation of Recognition, Diagnosis, Decision-Making, Planning, and Execution

Agricultural tasks are highly composite and continuous; hence, dynamic task orchestration is required after task triggering. Task orchestration refers to breaking down a high-level business goal into several executable sub-tasks and invoking agents according to their dependency relationships. Taking pest and disease control as an example, the high-level goal is to "control disease spread and reduce yield loss," but the actual process requires sequentially completing anomaly recognition, disease diagnosis, risk assessment, knowledge retrieval, prevention and control decision-making, path planning, spray execution, and re-examination feedback. Different sub-tasks correspond to different model capabilities and agent roles; only by completing dynamic orchestration can the agricultural AI agent transcend a simple Q&A system to become a workflow system capable of organizing agricultural production tasks.

Taking greenhouse tomato disease control as an example, the following business workflow can be constructed, as illustrated in Figure 6: First, the perception agent detects leaf or fruit anomalies via cameras, while the Environment perception agent reads temperature, humidity, light, and ventilation states. Next, the diagnosis agent judges the disease type and severity based on the image model output, environmental parameters, and disease knowledge base. Then, the knowledge agent retrieves disease control knowledge, historical treatment cases, and pesticide usage rules, and, based on this, the decision agent generates spraying, ventilation, dehumidification, or isolation plans. Subsequently, the planning agent generates inspection or spraying paths according to greenhouse layout, robot positions, and diseased plant distribution, and the execution agent invokes sprayers, ventilation equipment, or irrigation systems to complete the operation. Finally, the feedback agent judges the control effect based on re-examination images, environmental data, and operation records and updates the knowledge base.

This task orchestration logic is also applicable to smart irrigation and agricultural machinery operations. Smart irrigation tasks can be decomposed into data collection, water requirement estimation, irrigation decision-making, valve control, and effect evaluation; an irrigation decision review points out that irrigation systems need to integrate weather, soil, crops, and control strategies to determine irrigation timing and volume [66]. Agricultural machinery operations can be decomposed into environmental mapping, task area division, path planning, trajectory tracking, obstacle avoidance, and multi-machine collaboration; orchard robot path planning research proposes hybrid path planning algorithms addressing narrow orchard aisles, dense trees, and navigation safety, providing path planning capabilities for Execution Agents [122]. A multi-scenario autonomous navigation review further notes that future agricultural robots need to adapt to diverse scenes like open fields, orchards, and greenhouses and achieve autonomous task matching and multi-machine collaboration [123]. However, the transferability of a workflow across agricultural businesses should not be overestimated. Disease control workflows are usually diagnosis-centered, irrigation workflows are prediction- and control-centered, and machinery operation workflows are planning- and safety-centered. These differences imply that a reusable agricultural AI agent workflow should retain a common organizational logic while allowing task-specific capability combinations. Current studies still lack sufficient validation of such cross-business transferability. In particular, there is limited evidence showing whether an agent orchestration mechanism designed for one scenario can be reliably transferred to another scenario without redefining task semantics, capability boundaries, and safety constraints.

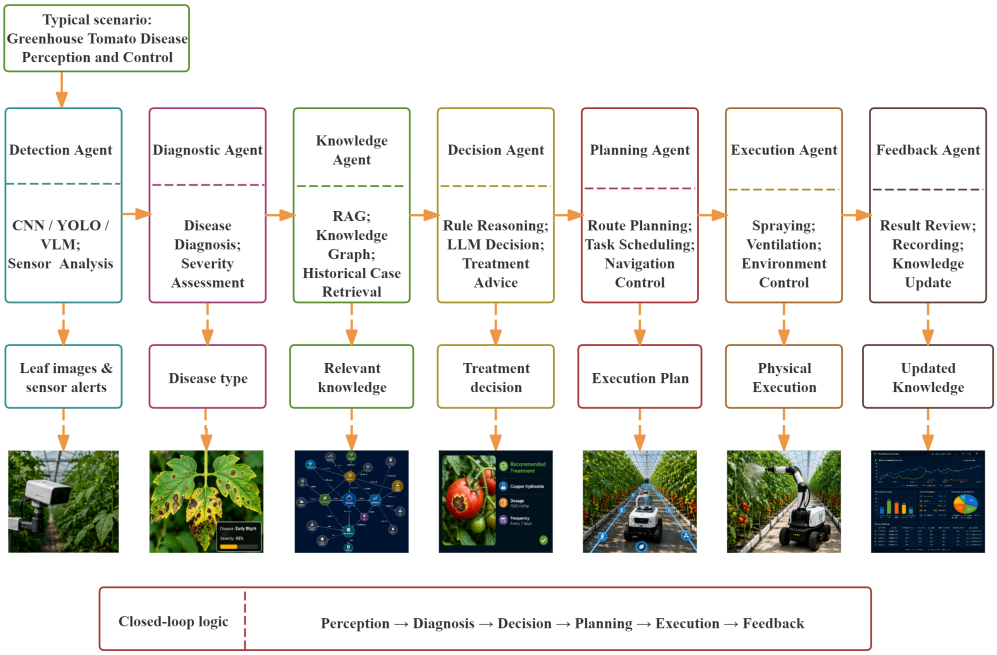


Figure 6. Greenhouse tomato disease control agent workflow.

4.5. Collaborative Execution: From Intelligent Decision-Making to Physical Agricultural Operations

Collaborative execution is the critical step for agricultural AI agents to move from digital decision-making to physical operations, distinguishing them from general large model Q&A systems. In actual agricultural operations, the solutions provided by agents typically need to be executed by physical equipment such as sprayers, irrigation systems, ventilation equipment, mobile robots, drones, tractors, or implements. Therefore, agricultural AI agent workflows must include software–hardware interfaces and execution coordination mechanisms so that decision results can be translated into operable agricultural actions on-site.

In precision plant protection operations, image recognition results can directly drive variable-rate spraying. Research on targeted weed control in strawberries developed a deep learning-based variable pesticide spraying system, enabling spraying equipment to perform spot-specific operations based on weed recognition results, thereby reducing chemical input and environmental pollution risks [124]. A review of intelligent orchard spraying technology shows that orchard spraying is shifting from traditional constant-rate application to intelligent spraying based on canopy perception, target recognition, and variable-rate control [36]. Related research demonstrates that execution agents do not simply execute "on/off" commands, but must complete fine control by integrating target locations, crop status, equipment posture, and operating parameters.

In water and facility environment management tasks, execution agents must coordinate irrigation, ventilation, heating, shading, and supplemental lighting equipment. A smart sprinkler irrigation review emphasizes that remote autonomous sprinkler irrigation relies on the joint support of soil moisture sensors, wireless sensor networks, communication systems, remote management, and execution control [121]. A greenhouse intelligence review further summarizes the roles of PID control, model predictive control, machine learning, and reinforcement learning in greenhouse environment regulation [25]. If these technologies are placed into an agricultural AI agent workflow, they can be understood as collaborative interfaces for "Decision Agents—Control Agents—Execution Equipment."

In agricultural machinery and robot operations, collaborative execution also involves path coordination, operation safety, and multi-machine cooperation. A review of AI-driven tractor-implement collaborative control notes that autonomous agricultural equipment is evolving from single-machine automation to multi-agent collaborative control systems, with key issues including real-time collaborative planning, perceptual robustness under environmental disturbances, and safety control under operational disturbances [37]. An AI in agricultural equipment review also shows that the integration of multi-modal perception, intelligent control, and agricultural equipment is pushing agricultural machinery towards intelligent, automated, and sustainable directions [2]. Therefore, the core of the collaborative execution layer is enabling agricultural AI agents to connect with physical equipment and achieve safe, precise, and schedulable physical operations in complex agricultural environments.

4.6. Result Feedback and Knowledge Update: From Single Tasks to a Continuous Optimization Loop

Result feedback and knowledge updating are key to closing the loop in the agricultural AI agent workflow. Traditional agricultural automation systems often complete a control task based on preset rules, whereas agricultural AI agents should re-collect images, sensor, and operational data post-execution to judge whether the task achieved the expected effect and write the results into knowledge bases, operational archives, or model update mechanisms. Taking disease control as an example, after spraying or ventilation, the system needs to re-detect whether lesions have expanded, environmental humidity has dropped, pesticide coverage is sufficient, and plant health has recovered. If the control effect is inadequate, the feedback agent should re-trigger the diagnosis and decision-making process.

Agricultural knowledge bases, knowledge graphs, and RAG technologies provide important support for feedback updates. Research on smart agriculture knowledge bases proposes organizing knowledge on crop production, agricultural resources, machinery, and precision tasks based on ontological principles, allowing production knowledge to be stored, edited, validated, and visualized [53]. The existing literature indicates that feedback data should not only be saved as historical records but should be further transformed into experiential knowledge that agents can retrieve and invoke. More specifically, the feedback agent can be organized as a three-stage mechanism of "recording–validation–knowledge updating." In the recording stage, the agent stores the task objective, input data, selected agents, decision path, execution parameters, environmental context, and post-operation outcomes as a structured operational record. In the validation stage, the agent compares the expected outcome with post-execution observations, such as lesion changes, humidity reduction, pesticide coverage, irrigation response, or equipment trajectory deviation. Only validated feedback should be converted into reusable knowledge. In the knowledge updating stage, operational records can be transformed into new cases, revised rules, or graph triples, such as "disease type–control measure–observed effect," "environmental condition–operation parameter–crop response," or "equipment state–execution deviation–correction strategy." In this way, feedback is not merely stored as a log but becomes retrievable experience for future diagnosis, planning, and decision-making. To avoid introducing bias during autonomous knowledge updating, feedback data should not be directly written into the core knowledge base without verification. Several safeguards are necessary. First, abnormal or low-confidence observations should be marked as provisional records rather than confirmed knowledge. Second, feedback from different plots, seasons, crop varieties, and equipment types should be stored with contextual metadata to prevent overgeneralization from a single scenario. Third, conflicting feedback should trigger human expert review or multi-source evidence comparison before being used to update

knowledge graphs or retrieval databases. Fourth, the system should retain versioned knowledge records so that incorrect updates can be traced, rolled back, or corrected. These mechanisms can help agricultural AI agents balance autonomous learning with agronomic reliability, preventing short-term operational noise from becoming long-term decision bias.

Throughout the entire agricultural production process, feedback mechanisms are also closely related to safety traceability and trusted management. Multi-agent-based agricultural product quality safety traceability systems achieve quality safety early warning, supervision control, and system evaluation through multi-agent division of labor, providing references for archive recording and traceability management during production [92]. Cloud-edge collaborative and digital twin architectures can provide real-time computing, low-latency control, and virtual–real mapping capabilities for agricultural AI agents, allowing feedback data to flow among the cloud, edge, and equipment ends [51]. Thus, the complete form of an agricultural AI agent should be a verifiable and updatable closed-loop system of "task triggering—capability matching—agent orchestration—collaborative execution—feedback validation—knowledge updating," rather than a purely linear "input–output" system. The effectiveness of this loop depends not only on whether feedback is collected but also on whether feedback is validated, contextualized, and transformed into reusable knowledge without reducing system reliability.

4.7. Summary: A Reusable Workflow Paradigm for All Agricultural Businesses

In summary, agricultural AI agent workflows should be constructed based on the model capabilities required for typical tasks. Operations such as pest and disease control, smart irrigation, greenhouse regulation, agricultural machinery operations, and agricultural knowledge services, despite having different task targets, can all be abstracted into a general process of "business requirement identification—model capability matching—agent selection—task decomposition and orchestration—collaborative execution—result feedback and knowledge update." This process not only covers the perception, diagnosis, decision-making, planning, and execution stages in agricultural production but also forms continuous optimization through knowledge bases and feedback mechanisms.

Compared with reviews of single-point agricultural AI technologies, the value of the workflow perspective lies in reorganizing scattered research—such as object detection, disease diagnosis, irrigation decision-making, greenhouse control, path planning, variable-rate spraying, machinery collaboration, and knowledge services—into a closed agricultural business loop. Nevertheless, the current workflow paradigm remains more mature as an analytical framework than as a fully validated engineering architecture. Existing studies provide strong evidence for individual capabilities such as perception, irrigation decision-making, path planning, and variable-rate spraying, but fewer studies evaluate how these capabilities behave when chained into long-horizon agent workflows. Future research should therefore move from single-module performance evaluation toward workflow-level validation, including task success rate, failure recovery, human intervention frequency, communication cost, execution safety, and knowledge-update reliability. Such evaluation is necessary for determining whether agricultural AI agents can operate robustly beyond controlled demonstrations and across diverse agricultural scenarios. The literature from Jiangsu University and related agricultural engineering fields provides substantial underlying support for the perception, decision-making, planning, and execution capabilities therein; meanwhile, literature on agricultural AI agents, RAG, large models, and multi-agent systems provides framework support for upper-level task organization, knowledge retrieval, and collaborative decision-making.

5. Key Technologies of Agricultural AI Agents

Agricultural AI agents have become important enablers for the transition of smart agriculture from single-point automation to autonomous execution of complex tasks. Existing studies show that these agents need to integrate multimodal perception, knowledge modeling, intelligent decision-making, and closed-loop optimization and can be applied to crop production, resource use optimization, agricultural machinery upgrading, and value-chain governance [125]. With the convergence of large language models (LLMs) and multi-agent systems (MASs), agricultural AI agents are evolving from isolated perception or control modules into comprehensive intelligent units with role specialization, real-time perception, collaborative decision-making, and closed-loop execution [15]. Meanwhile, studies on agricultural multi-robot systems, precision irrigation agents, and heterogeneous equipment collaboration show that agricultural tasks are characterized by dynamic environments, long task chains, heterogeneous collaborators, and resource constraints. Therefore, the key technologies of agricultural AI agents should be examined from target understanding and agent selection, multi-agent collaboration and consensus, and multi-agent scheduling and workflow orchestration [96,126]. Table 3 summarizes these technologies and their integration methods in practical scenarios.

Table 3. Summary of key technologies and their advantages and limitations.

Domain	Key Technology	Advantages	Limitations	References
Target Understanding and Agent Selection	Natural Language Understanding, Vision, Task-based Selection	Enables task goal understanding, visual scene reasoning	Precision issues, inconsistent in complex environments	[1,4]
Multi-agent Cooperation and Consensus Mechanism	Consistency Control, Cooperative Mechanisms, Multi-agent Learning	Ensures task coordination and stability	High computation and communication costs	[9,12]
Multi-agent Scheduling and Workflow Orchestration	Path Planning, Task Assignment, Workflow Coordination	Optimizes resource use and task allocation	High complexity under dynamic conditions	[2,3,5]

5.1. Goal Understanding and Agent Selection

Whether an agricultural AI agent system can accomplish complex tasks depends on its ability to understand task goals and select suitable agents, equipment, or toolchains. Agricultural tasks are highly context-dependent, involving factors such as plot location, crop growth status, equipment condition, operation windows, and risk constraints. Therefore, goal understanding and agent selection require converting language instructions, visual observations, sensor signals, and historical experience into computable task representations, followed by capability matching and execution entity selection. Existing studies suggest that intent understanding, agent modeling, and agent selection are key prerequisites for reliable collaboration, scheduling, and execution [15,56].

To summarize the relationship between these tasks and the required model capabilities, Table 4 outlines the key technologies and their respective advantages and limitations in different contexts.

Table 4. Key technologies for goal understanding and agent selection in agricultural AI agents.

Category	Technology	Advantages	Disadvantages	References
Intent Understanding	Scene Perception, Language Conversion, Hierarchical Planning	Suitable for complex tasks, lowers automation barrier	Dependent on recognition quality, complex systems	[56,127,128]
Agent Modeling	Task Complexity, Multi-modal Perception, Knowledge Graphs	Clarifies task difficulty, enhances accuracy	Evaluation errors, hardware needs, data inconsistency	[15,56]
Agent Selection	Task-based Selection, Adaptive Scheduling, Multi-Agent Cooperation	Matches agents, dynamic adjustment, boosts efficiency	Evaluation errors, unpredictable results, high cost	[9,12,127]

5.1.1. Intent Understanding

Existing research identifies three main approaches in agricultural intent understanding. One approach begins with scene perception, where scene structures are built through object detection, relationship recognition, and attribute extraction and then mapped to operational goals. The visual scene-understanding task-planning framework proposed by Park et al. exemplifies this approach. The system first identifies fruits, branches, and leaves along with their relationships and attributes using scene graph recognition. It then triggers task decisions, such as picking, pruning, or thinning, and, finally, determines whether single-arm or dual-arm execution should be adopted [10]. The main advantage of this method lies in its ability to directly connect the "perception–understanding–decision" chain, making it particularly suitable for embodied agricultural robots. However, its limitations are also clear: it is highly sensitive to perception quality, and front-end recognition errors can easily be amplified during task discrimination and action selection, especially in complex execution scenarios such as dual-arm collaboration. Compared with language-driven approaches, scene-perception-based methods are more tightly coupled with embodied agricultural operations, but their performance remains highly dependent on perception robustness under complex field conditions [10,12]. In agricultural environments with occlusion, illumination variation, and dense crop structures, perception errors can easily propagate through the subsequent reasoning and planning chain, thereby reducing execution reliability.

A second line of research places greater emphasis on converting natural language into intermediate task representations. With the development of large language models, increasing attention has been devoted to transforming agricultural natural language instructions into structured task descriptions. The heterogeneous agricultural robot natural language task-planning framework proposed by Zuzu'arregui et al. uses large language models and predefined operational primitives to translate commands from non-expert users into intermediate representations that can be executed across different robotic platforms. This approach enables no-code planning and the execution of complex agricultural tasks [11]. This approach significantly lowers the entry barrier for agricultural automation systems and offers clear advantages in terms of user interaction and cross-platform generality. However, it also imposes higher requirements on domain knowledge constraints, ambiguity resolution, and result verifiability. Without agricultural knowledge enhancement and execution feedback loops, the system is prone to semantic drift, omitted steps, or inconsistencies with real agricultural constraints. Although LLM-based methods improve interaction flexibility and reduce operational barriers for non-expert users, they also introduce challenges related to hallucination, ambiguity resolution, and agricultural knowledge consistency [7,11]. Compared with perception-driven approaches, language-driven systems exhibit stronger

generalization capability in open interaction scenarios, but their dependence on external knowledge constraints and verification mechanisms remains significantly higher [7,11].

A third line of work embeds intent understanding into the process of task complexity assessment and planning activation. AgriAgent is representative of this approach: it divides agricultural tasks into simple and complex tasks, with the former handled directly by modality-specific agents and the latter transferred to contract-driven planning based on capability requirements, supported by capability-aware tool orchestration, dynamic tool generation, and failure recovery for multi-step execution [125]. Compared with the previous two paths, this hierarchical form of intent understanding is more consistent with complex task chains in real agricultural scenarios because it unifies "understanding the goal" with determining whether planning is necessary, whether tools need to be supplemented, and whether failure recovery is required. Correspondingly, however, it comes at a higher cost: the system architecture is more complex and it depends more heavily on the completeness of capability descriptions and the tool ecosystem.

Overall, agricultural intent understanding has moved beyond the simple identification of input content and is gradually shifting toward execution-oriented task semantic modeling. Scene-perception-based methods are more suitable for embodied operational systems, language-driven methods are more appropriate for interactive agricultural applications, and hierarchical planning methods are better aligned with complex, multi-step, and open-environment tasks. This comparison suggests that current agricultural intent-understanding systems are gradually evolving from isolated perception or language parsing toward integrated semantic reasoning and execution-oriented planning [7,10,125]. However, existing studies still lack unified evaluation benchmarks for intent understanding under dynamic agricultural environments, particularly in scenarios involving multimodal interaction, uncertain environmental conditions, and long-term task continuity [1,9]. The key challenge for future research is not to replace one path with another but to integrate multimodal understanding, task semantic abstraction, and planning linkage under agricultural knowledge constraints, thereby improving the stability and interpretability of agricultural agents in handling real-world complex tasks.

5.1.2. Agent Modeling

Existing research has formed three representative approaches to agent modeling. The ontology-driven approach emphasizes unified semantic representation. Ontologies can organize agricultural tasks, equipment capabilities, and control processes into a unified semantic structure, thereby providing a foundation for information exchange and collaborative control among agricultural devices [129]. The semantic-knowledge-driven approach further links agricultural environments, service capabilities, and task conditions, enabling systems to perform environmental reasoning and goal planning in the presence of heterogeneous equipment and multiple service providers [54]. In contrast, the dynamic capability profiling approach represented by OpenAg places greater emphasis on integrating foundation models, knowledge graphs, multi-agent reasoning, and causal explainability within a unified agricultural knowledge base, thus producing capability descriptions that can adapt to changing scenarios [130]. The first two approaches are advantageous in terms of standardized representation and interoperability, whereas the latter is more consistent with the practical reality that capability boundaries and resource constraints in agricultural scenarios may vary across tasks and environments.

This comparison suggests that current agricultural AI agent modeling is gradually shifting from static semantic representation toward adaptive and context-aware capability modeling. Ontology-driven approaches are beneficial for interoperability and semantic consistency, while dynamic capability profiling methods are more suitable for highly

variable and open agricultural environments [129,130]. However, the trade-off between standardized representation and dynamic adaptability remains insufficiently resolved in existing studies, particularly in scenarios involving heterogeneous equipment, changing environmental conditions, and long-term collaborative operations [125].

Nevertheless, several limitations remain in current agricultural agent modeling research. First, unified descriptions of composite agricultural AI agent systems are still insufficient, as many studies focus primarily on the capability representation of a single device or service. Second, different studies vary widely in how they characterize capability boundaries, resource constraints, and applicable conditions, making direct reuse difficult. Third, in human–machine collaborative scenarios, human experts have not yet been fully incorporated into a unified capability representation framework [54,125]. As a result, agent selection is not a straightforward invocation following modeling but rather a dynamic configuration process that depends heavily on the quality of the modeling itself.

5.1.3. Agent Selection

Agent selection is the process of dynamically matching appropriate agents or tools according to task objectives, environmental conditions, and resource constraints. Early agricultural AI agent selection strategies mainly relied on static rule matching or predefined service mappings, which were suitable for relatively simple and deterministic agricultural workflows. However, as agricultural tasks become increasingly complex and open-ended, recent studies have gradually shifted toward capability-aware and collaborative selection mechanisms.

One representative direction focuses on capability-driven dynamic orchestration. OpenAg integrates large language models, knowledge graphs, and multi-agent coordination mechanisms to dynamically organize agricultural AI agent tools and services according to task requirements and environmental states, thereby improving adaptability under heterogeneous agricultural scenarios [130]. Another line of work emphasizes contract-based and hierarchical planning strategies. AgriAgent divides agricultural tasks into simple and complex categories, allowing lightweight tasks to be executed directly while transferring complex tasks to collaborative planning modules involving tool generation, failure recovery, and dynamic orchestration [125]. Compared with traditional static assignment strategies, these approaches increasingly emphasize dynamic capability matching and temporary collaborative structures, reflecting the growing complexity and uncertainty of real agricultural task environments.

Capability-aware selection mechanisms are more suitable for heterogeneous agricultural systems because they can dynamically adapt to changing task requirements and environmental conditions [125,130]. However, most current selection frameworks still rely heavily on predefined capability descriptions and relatively stable execution assumptions, limiting their robustness under highly dynamic field conditions. In addition, current studies rarely consider long-term collaboration stability, resource competition, and human–agent mixed participation simultaneously, leaving practical large-scale deployment insufficiently explored.

Overall, agricultural agent selection is gradually evolving from static invocation mechanisms toward adaptive collaborative orchestration. The key challenge is no longer simply selecting "which agent to use" but rather dynamically organizing heterogeneous agents, tools, and human participants under uncertain agricultural environments. Future research should therefore further strengthen real-time capability assessment, adaptive coordination, and human–agent collaborative selection mechanisms to improve the robustness and scalability of agricultural intelligent-agent systems.

5.2. Multi-Agent Collaboration and Consensus Mechanisms

The key challenge in agricultural AI agent systems is not only whether a single agent can perform localized perception or reasoning but whether multiple agents can collaborate reliably around the same agricultural task. Agricultural scenarios involve dynamic environments, heterogeneous actors, and high operational risks. Collaborative entities may include UAVs, ground robots, irrigation controllers, sensing nodes, knowledge-service modules, reasoning agents, and human managers. Weather changes, equipment failures, unstable networks, and inter-plot differences further increase coordination difficulty. Therefore, practical multi-agent systems require more than simple coexistence; they need unified interaction rules, conflict resolution mechanisms, and trustworthy long-term collaboration. This subsection focuses on three prerequisite aspects of collaboration: protocol communication, conflict resolution, and trust management [15,125,126]. Table 5 summarizes the key technologies, advantages, and limitations of these aspects in agricultural multi-agent systems.

Table 5. Key technologies in multi-agent collaboration, conflict resolution, and trust management.

Technology Category	Technology	Advantages	Limitations	References
Protocol Communication	Ontologies, Knowledge Graphs	Facilitates semantic exchange and system interoperability	Increases knowledge engineering costs, data inconsistency issues	[15,96]
Conflict Resolution	Constraint Optimization, Task Allocation, Path Generation	Resolves path collisions and task coordination	High computational complexity, difficult to adapt in dynamic environments	[126,131]
Trust Management	Federated Learning, Explainability, Cybersecurity	Improves data quality, enhances system transparency	Affected by data heterogeneity, device computing power differences	[132,133]

5.2.1. Protocol Communication

Protocol communication is particularly important in agricultural multi-agent systems because agricultural collaboration is never merely about whether messages can be transmitted but rather about whether different entities can continuously share semantics, context, and state information around the same task. This issue is especially prominent in agricultural scenarios, where systems often connect devices from different manufacturers, protocol stacks, and computing platforms while simultaneously facing practical constraints such as limited edge computing resources, unstable network coverage, and highly heterogeneous data formats. Therefore, the core difficulty of agricultural multi-agent communication lies less in connectivity itself than in interoperability [15,96].

Existing studies indicate that communication mechanisms for collaborative agricultural equipment must first address the issue of semantic alignment. Ontologies and knowledge graphs have been repeatedly adopted in smart agriculture because they can organize agronomic tasks, device capabilities, environmental states, and control processes within a unified semantic framework, enabling different entities to exchange not merely raw data but information endowed with explicit meaning and operational constraints [96,125]. This is particularly critical in agricultural settings: if an irrigation agent, an environmental monitoring agent, and a decision-service module interpret the state of "insufficient soil moisture" differently, subsequent coordinated control can easily become distorted. Although unified semantic description increases the cost of knowledge engineering, it also provides a relatively stable common language for subsequent collaboration.

With the development of large models and multi-agent frameworks, the focus of agricultural communication has further shifted from device interoperability to task interoperability. Relevant reviews have pointed out that current multi-agent, large-model systems increasingly emphasize inter-agent communication, communication protocol frameworks, and communication security. Protocols such as MCP, A2A, ANP, and Agora essentially attempt to answer the same question: when different agents possess different capabilities, contextual information, and responsibilities, how can they exchange tasks, synchronize states, and sustain collaboration [15]? This perspective is highly instructive for agricultural scenarios because agricultural task chains are often long. From data collection, information cleaning, and knowledge reasoning to equipment execution and feedback optimization, agricultural tasks can rarely be completed through a single round of interaction. What agricultural multi-agent collaboration truly requires is a communication mechanism capable of carrying capability descriptions, task invocation, and state transition, rather than merely a low-level message transmission channel.

Furthermore, effective communication in agricultural scenarios is inherently task-oriented. Precision irrigation research has shown that what agents exchange are not abstract messages but state information, environmental variables, and decision outputs directly related to when to irrigate, how much water to apply, and whether strategies need to be adjusted [96]. Similarly, in agricultural decision-support systems such as AgroAskAI, multiple role-specialized agents iteratively pass queries, supplement external evidence, and revise intermediate judgments through a chain-of-responsibility mechanism, such that communication itself becomes embedded in the reasoning process [134]. Interactions among the Retriever, Reflector, Answerer, and Improver in multi-image agricultural question answering likewise demonstrate that high-quality agricultural reasoning often depends on a continuous information flow of "retrieve–reflect–generate–revise," rather than single-round, static, and feedback-free message exchange [135]. In this sense, as agricultural multi-agent communication continues to evolve, it becomes less like traditional protocol adaptation and more like a form of semantic coordination organized around task goals.

Nevertheless, several limitations remain. A stable mapping has not yet been established between agricultural device ecosystems and general-purpose, multi-agent protocol frameworks, leaving many systems with weak transferability. Under edge–cloud collaboration and multi-round reasoning, state drift and contextual loss remain widespread. Moreover, although communication security is increasingly discussed, protocol-level security designs specifically targeting high-risk agricultural scenarios remain relatively underdeveloped [15,133]. Therefore, future breakthroughs in agricultural protocol communication should not focus solely on transmitting information faster but also on aligning it more robustly, sharing it more securely, and embedding it more naturally into collaborative processes.

5.2.2. Conflict Resolution

Conflict resolution is a core issue in agricultural multi-agent collaboration because agricultural environments often involve resource competition, inconsistent observations, task conflicts, and dynamic environmental uncertainty. Early agricultural conflict resolution strategies mainly focused on physical resource coordination and rule-based scheduling, such as avoiding machinery collisions, resolving task allocation conflicts, and balancing resource consumption. Although these approaches are effective under relatively deterministic environments, their adaptability becomes limited as agricultural systems gradually evolve toward open, heterogeneous, and knowledge-intensive collaborative environments.

Recent studies have increasingly introduced semantic reasoning and reflective coordination mechanisms into agricultural conflict resolution. Knowledge-driven coordination frameworks can resolve conflicts by incorporating environmental semantics, operational constraints, and capability relationships into collaborative reasoning processes [54]. Meanwhile, large-language-model-based agricultural agents further extend conflict resolution from physical coordination toward cognitive consistency management. OpenAg integrates foundation models, causal reasoning, and multi-agent collaboration mechanisms to support adaptive negotiation and dynamic decision adjustment under uncertain agricultural environments [130]. Compared with traditional optimization-based approaches, reflective reasoning frameworks are more suitable for knowledge-intensive agricultural tasks because they can dynamically revise decisions according to contextual feedback and semantic inconsistencies.

However, these emerging approaches also introduce new challenges. Large-model-driven coordination mechanisms generally require higher reasoning costs, stronger contextual consistency, and more reliable verification mechanisms, while semantic-based coordination frameworks still depend heavily on the completeness and accuracy of agricultural knowledge representation [7,11]. This indicates that current agricultural conflict resolution mechanisms are gradually evolving from physical resource coordination toward cognitive and evidential consistency management. Nevertheless, their scalability, reasoning efficiency, and long-term operational stability under large-scale agricultural deployments remain insufficiently validated [125].

Additionally, current studies still lack unified evaluation frameworks for measuring conflict resolution effectiveness under dynamic agricultural environments involving heterogeneous agents, uncertain field conditions, and long-term collaborative tasks. This limitation restricts the comparability and practical transferability of many existing agricultural multi-agent systems.

Overall, agricultural conflict resolution is gradually shifting from static optimization and rule scheduling toward adaptive semantic coordination and reflective reasoning. Future research should therefore further strengthen scalable collaborative reasoning, trustworthy decision verification, and human–agent cooperative conflict management under real agricultural production environments.

5.2.3. Trust Management

If protocol communication addresses whether agents can exchange information smoothly, and conflict resolution addresses how disagreements can be transformed into coordinated action, then trust management confronts a deeper issue: in the course of sustained collaboration, whom should the system trust, on what basis should that trust be established, and how should it be adjusted once a certain entity or type of information is no longer reliable? This issue is particularly critical in agricultural scenarios because the outputs of agricultural agents often directly affect irrigation, pest and disease control, scheduling, and operational execution. Once erroneous judgments are amplified along the collaborative chain, the consequences may include not only reduced efficiency but also resource waste, risk propagation, and even operational failures [54,136].

Existing research most relevant to trust management in agricultural agent systems is mainly concentrated at three levels. The first concerns whether the data source is trustworthy. As agricultural digitalization deepens, data collaboration across farms, organizations, and platforms is becoming increasingly common. Yet this collaboration is not merely a question of whether data can be shared but also whether the shared data remain worthy of reliance afterward. Federated learning is widely regarded as an important direction for agricultural privacy protection because it allows multiple participants to collaborate on

modeling without exposing raw data. However, relevant reviews have also pointed out that data heterogeneity, communication overhead, disparities in device computing power, and unequal benefit distribution can all undermine participants' trust in the system [54]. This suggests that in agricultural multi-agent collaboration, trust management is not an abstract attitude problem but rather an operational condition directly tied to data quality, data ownership, and collaborative fairness.

The second level concerns whether reasoning outcomes are trustworthy. For agricultural agent systems that increasingly rely on large models and deep learning models, users are often not satisfied merely by the fact that the agricultural agent has produced an answer, but are more concerned with how that answer was obtained, under which circumstances it applies, and under what conditions it may fail. Barbedo et al., in their review of soybean monitoring, argued that explainability and privacy protection should not be treated as auxiliary functions added after deployment but should instead be regarded as core design objectives from the outset [137]. Tsoumas et al. further emphasized that in digital agriculture, causality and explainability are of central importance for improving model transparency and user trust [138]. In multi-agent collaboration, this issue becomes even more complex because what must be trusted is no longer just the output of a single model but the intermediate judgments, information flows, and final decisions of the entire collaborative chain. The internal feedback, reflection, and revision mechanisms introduced in systems such as AgroAskAI and multi-image agricultural question answering already embody a form of trust enhancement oriented toward the collaborative chain: rather than assuming that an agent's first output is reliable, they improve the acceptability of results through multi-role scrutiny and iterative revision [134,135].

Compared with traditional security-oriented trust mechanisms, recent agricultural trust management frameworks increasingly emphasize collaborative trustworthiness involving data reliability, reasoning transparency, and system-level coordination. Federated-learning-based approaches mainly focus on protecting data privacy and collaborative fairness, whereas explainability-driven frameworks place greater emphasis on improving the interpretability and acceptability of reasoning outcomes [54,137,138]. In contrast, collaborative-chain-oriented systems such as AgroAskAI further extend trust management toward multi-role verification and iterative reasoning consistency [134,135]. This evolution indicates that agricultural trust management is gradually shifting from isolated security protection toward integrated collaborative trust governance.

The third level concerns whether the collaborative system itself is trustworthy. Agricultural agents do not operate in isolation; instead, they are embedded in complex infrastructures composed of sensor networks, edge nodes, cloud platforms, and control devices. Therefore, even if a particular agent produces a correct reasoning outcome, the collaborative result may still fail if the perception chain, communication chain, or platform environment is fragile. Reviews on smart agriculture cybersecurity have already shown that cyberattacks, device tampering, and systemic vulnerabilities are becoming key factors affecting the reliability of agricultural digital systems [139]. This means that trust management in multi-agent systems cannot remain confined to the model layer or the data layer but must be extended to a more complete system governance layer, including source authentication, anomaly detection, responsibility tracing, and the isolation of untrustworthy nodes.

However, current discussions of trust management in agriculture remain rather fragmented. Many studies separately address privacy, security, or explainability, yet comparatively few begin from the collaborative relations themselves and systematically explain how dynamic trust can be established among different entities, how trustworthiness can be adjusted through long-term interaction, and how collaborative stability can be maintained when abnormal entities arise. Nevertheless, current agricultural trust management studies

still lack unified quantitative evaluation frameworks for measuring long-term trust stability under heterogeneous agricultural collaboration environments. This indicates that trust management for agricultural agents cannot remain limited to protecting data or explaining results but must gradually move toward a more comprehensive mechanism oriented toward the collaborative chain, one that truly connects data trustworthiness, reasoning trustworthiness, and system trustworthiness. Only in this way can multi-agent collaboration become not merely a one-time coordination strategy but a sustainable and dependable mode of agricultural intelligent operation [54,136].

5.3. Multi-Agent Scheduling and Workflow Orchestration

If the previous section discussed how agricultural multi-agent systems establish collaborative relationships, this section focuses on how these relationships evolve into continuous, executable, and adaptive operational states. Agricultural tasks are rarely static or one-off; instead, they consist of multiple interdependent stages. Perception outputs must be communicated to analytical modules, analytical conclusions must inform decision generation, and decisions must subsequently drive equipment execution and feedback correction. Therefore, the challenge for agricultural multi-agent systems lies not only in having multiple agents but in organizing them into a task chain that can operate continuously, adapt to environmental changes, and maintain stability throughout execution. Based on this, the present subsection examines three key operational issues for agricultural multi-agent systems: workflow orchestration, resource scheduling, and state synchronization [56,135,140].

Although these three issues are often discussed separately, they are in fact tightly coupled in real agricultural scenarios. Workflow orchestration determines the order in which the task chain unfolds, resource scheduling determines which entities receive priority support and when, and state synchronization determines whether the agricultural multi-agent system can continuously track its current progress and perceive changes in the external environment. Together, these three aspects constitute the executional foundation through which agricultural multi-agent systems move from being merely operational to being stably operational. These core issues and their interdependencies are summarized in Table 6.

Table 6. Key technologies in multi-agent scheduling and task allocation.

Category	Technology	Advantages	Limitations	References
Workflow Orchestration	Task chain orchestration, reflective process chain	Adaptable to changes, continuous task execution	Limited by incomplete tools, scenario-specific designs	[56,135]
Resource Scheduling	Collaborative scheduling, cloud–edge–device integration	Improves efficiency and resource management	Struggles with environmental variability, mixed scenarios	[62,63,141]
State Synchronization	Digital twin, process-state consistency	Ensures real-time adaptation and coherence	Data inconsistency, wireless synchronization challenges	[60,142]

5.3.1. Workflow Orchestration

For agricultural multi-agent systems, workflow orchestration is not simply about listing tasks in sequence but about organizing multiple subtasks into a task chain that can be executed, rolled back when necessary, and adaptively adjusted in response to environmental changes. Agriculture relies particularly heavily on workflow orchestration

because agricultural tasks typically exhibit a clear chain structure. For example, in a pest and disease control task, the agent workflow often needs to complete multisource data collection first, then conduct anomaly detection and diagnostic analysis, subsequently generate control recommendations, and, finally, hand them over either to execution devices for implementation or to human operators for confirmation. Without a unified orchestration mechanism, these steps may each be individually feasible yet difficult to connect into a stable and complete closed loop [135,140].

Existing studies have begun to address this issue from multiple perspectives. The value of AgriAgent lies in its hierarchical execution strategy, which adapts according to task complexity rather than forcing all agricultural tasks into a single execution paradigm. Simple tasks are completed directly by modality-specific agents, while complex tasks are transferred to contract-driven planning and accomplished through capability-aware tool orchestration, dynamic tool generation, and failure recovery mechanisms [56].

This indicates that agricultural workflow orchestration is evolving from a fixed pipeline toward a task-driven elastic process. In other words, the system no longer assumes that each task must follow the same path; instead, it can dynamically adjust the process structure based on task complexity and tool availability. Similarly, in agricultural multi-image question-answering systems, the Retriever, Reflector, Answerer, and Improver components form a reflective and iterative process chain rather than operating in parallel. Retrieval expands the context, reflection checks sufficiency, answering generates candidate results, and improvement corrects and strengthens existing results [135].

These examples suggest that the essence of agricultural workflow orchestration is not simply step-by-step execution. Rather, each step must proceed based on the context established by previous steps, with the flexibility to roll back or retry when necessary.

Workflow orchestration is equally important in agricultural physical operation scenarios, although there it is expressed more as an execution chain than as a reasoning chain. Santilli et al. pointed out in their framework for precise agricultural task allocation and scheduling that once there are priority constraints or sequential dependencies among agricultural operations, static task allocation alone is insufficient; the execution framework must also construct a complete execution sequence beyond simple allocation [143]. Li et al. further demonstrated that if task coordination and path generation are handled separately, the agricultural execution framework is prone to local optima but global inefficiency; accordingly, task allocation and spatial action generation should be considered jointly within the same framework [131]. Although such studies do not always explicitly employ the term "workflow orchestration," their focus is consistent with it: the key issue is not whether a particular action can be executed but how the entire operation chain can proceed more smoothly and efficiently.

Several notable weaknesses remain in current research on agricultural workflow orchestration. Many systems still favor single-scenario customization, which limits reusability and transferability. Many process designs assume relatively stable inputs and toolchains, whereas in real agricultural tasks, unavailable tools, missing data, and environmental changes are common, requiring stronger elasticity and recovery capabilities. In addition, a truly unified connection mechanism between high-level reasoning workflows and low-level embodied execution workflows is still lacking: many systems can arrange ideas effectively but fail to connect actions effectively [56,135,140]. Therefore, the most promising future direction for agricultural workflow orchestration is not to make processes more complex but to make them more adaptable to incomplete toolchains, more robust to mid-process failures, and more capable of establishing continuous mappings between knowledge reasoning and equipment execution.

5.3.2. Resource Scheduling

Once the workflow enters the operational stage, the focus shifts from simply determining "who is responsible for which step" to more practical concerns: which entities should receive priority support, when, and to what extent. At this level, resource scheduling becomes critical. Constraints in agricultural scenarios are more direct and complex than in many general systems. Resources include not only computing power, bandwidth, and storage but also electricity, UAV endurance, agricultural machinery operating windows, field traffic capacity, and even weather-dependent operational windows. In other words, agricultural resource scheduling is not merely a cloud computing problem; it is constrained by both the physical environment and digital systems [60,62].

In the context of agricultural robots and machinery collaboration, resource scheduling often involves a trade-off between task allocation and path costs. Cao et al. demonstrated that considering only the "proximity principle" can overload some equipment while leaving others idle, prolonging the operation cycle. Introducing a scheduling model that combines dynamic and static mechanisms can significantly improve both path costs and the overall operation cycle [141]. These findings indicate that agricultural resource scheduling is not a secondary concern but a core factor directly affecting collaborative efficiency. Li et al.'s integrated task and path planning framework further combines task coordination with trajectory generation, achieving a more balanced distribution between spatial movement cost and operational load [131]. These methods are applicable to multi-robot, multi-machinery, and multi-arm collaborative scenarios and primarily improve execution-level efficiency. However, most current approaches still favor single-task scenarios and offer limited support for more open or mixed agricultural tasks.

As smart agriculture systems increasingly move toward cloud–edge–device collaboration and IoT integration, the focus of resource scheduling has gradually expanded to include computing tasks, communication resources, and energy consumption management. Yu's review pointed out that cloud–edge–device collaborative computing is becoming increasingly important in agriculture precisely because traditional centralized architectures struggle to simultaneously satisfy real-time demands, bandwidth limitations, and distributed intelligence requirements, while agricultural scenarios inherently involve heterogeneous equipment, data consistency problems, resource shortages, and privacy constraints [60]. This indicates that in agricultural multi-agent systems, resource scheduling is expanding from "mechanical resource allocation" to the joint scheduling of physical resources and computing resources. Research by Tan et al., based on 6G edge computing and reinforcement learning, further shows that in the presence of dynamic environmental states and task demands, if resource allocation strategies cannot be updated adaptively, the execution framework will struggle to maintain a stable balance among latency, resource utilization, and energy efficiency [63]. Similarly, the GSAgri framework places task offloading, task scheduling, and energy prediction within the same collaborative MEC system, thereby achieving better overall performance in terms of latency, failure rate, and device energy consumption [62].

Resource scheduling in agricultural network environments also involves the joint coordination of communication links and mobile nodes. The UAV-assisted agricultural network model proposed by Xiong et al. models data packet scheduling and multi-hop routing simultaneously and then enhances transmission reliability and system robustness through a multi-agent reinforcement learning framework [144]. This work is particularly notable because it shows that in agricultural IoT environments, resource scheduling is not limited to allocating computing power or assigning operational tasks. It also requires dynamically balancing communication links, node positions, and task timing. Similarly, research on agricultural wireless sensing and edge systems indicates that system deployment is often

constrained not by the performance of a single algorithm but by the inability of multiple resources to simultaneously meet the demands of all entities [60,145].

A significant shift is thus taking place in agricultural multi-agent resource scheduling: it is evolving from task post-processing into a system precondition. In the past, many studies assumed that resources were sufficient and focused primarily on optimizing task outcomes; now, more and more work begins from the recognition that resources themselves are the primary constraint in the design of agricultural intelligent systems. This implies that the most promising direction for the future is not to optimize bandwidth, energy consumption, or routing in isolation but to construct a cross-layer resource scheduling mechanism that integrates operational tasks, computing tasks, communication tasks, and energy tasks into a unified framework for collaborative decision-making. Only in this way can agricultural multi-agent systems maintain both high efficiency and long-term operational capability in real and complex environments [60,62,63].

5.3.3. State Synchronization

In the overall execution mechanism, state synchronization is often not the most visible component, yet it frequently determines whether agricultural multi-agent framework can truly operate in a closed loop. Once multiple entities develop inconsistent perceptions of current progress, environmental changes, or task succession, even well-designed planning and scheduling mechanisms may fail during execution. When an agricultural multi-agent system enters the phase of continuous task execution, all participating entities must promptly know the current stage of progress, whether environmental conditions have changed, which nodes have completed their tasks, which nodes have failed, and which entity should take over next. If such state information cannot be shared reliably, the entire system may lose coordination because of information latency, even when task allocation is reasonable and resource scheduling is effective. This issue is particularly prominent in agricultural scenarios because environmental changes are highly time-sensitive, and crop conditions, soil moisture, equipment health, and network status may all change within a short period of time [60,145].

Existing research mainly presents two typical approaches to state synchronization. One is oriented toward task-process synchronization, that is, state updates are performed around the task chain itself. In complex task systems such as AgriAgent, multi-step execution, capability invocation, and failure recovery can be achieved only if the system always knows the current stage of the task, which tools have been successfully invoked, and which substeps need to be rolled back or retried [56]. The self-reflection framework in agricultural multi-image question answering is similar: the Retriever, Reflector, Answerer, and Improver can collaborate continuously because they share the same problem context and intermediate reasoning states; otherwise, subsequent improvements would lose their basis [135]. This type of state synchronization emphasizes process-state consistency and is more suitable for high-level reasoning and multi-round task execution.

The other approach is closer to synchronization between the physical world and the digital world. This direction is particularly evident in research on digital twins and real-time systems. Reviews of agricultural digital twins have pointed out that the key to a true digital twin lies not merely in having a virtual model but in whether real-time synchronization between the physical world and the digital world can be achieved; otherwise, the digital-twin framework is more like a static decision support tool than a continuously updated, collaborative system [142,146]. In crop digital twin systems, microclimate, crop growth, control strategies, and feedback results are continuously integrated into a closed loop, which in fact represents a higher level of state synchronization, not only among multiple agents but also between the system and the external world [146]. This type of method is

more suitable for continuous monitoring and adaptive control, but it also faces notable challenges: once the quality of multisource data fusion becomes unstable, time scales become inconsistent, or network latency becomes excessive, the so-called synchronization can easily degrade into low-frequency updating or even pseudo-synchronization.

Research on cloud–edge–end collaboration further indicates that agricultural state synchronization is not only a matter of data refresh frequency but also involves node distribution, data consistency, and system control logic. Yu et al. explicitly identified data consistency as a key challenge in agricultural cloud–edge–end collaborative architectures, highlighting the essence of state synchronization: whether states maintained across different computing layers remain consistent under multi-node collaboration directly affects the accuracy of subsequent decisions [60]. If edge nodes make control decisions based on outdated states while cloud nodes have already updated their risk assessments, the agricultural agent system may appear operational, but its collaborative integrity is compromised. Therefore, state synchronization should be treated not merely as an auxiliary communication function but as a core component of agricultural, multi-agent, closed-loop control.

Agricultural state synchronization is more difficult to handle than in many general systems because it concerns not only software states but also continuously changing physical environmental states. The cost of inconsistency across nodes is not limited to task delay; it may directly manifest as control mismatch, resource waste, and even equipment malfunction. Although existing studies have provided useful foundations in task-chain synchronization, digital twin synchronization, and cloud–edge–end consistency, several clear shortcomings remain. First, most systems still lack a unified state representation framework, making it difficult for different entities to share truly reusable state information. Second, the temporal consistency of multisource state updates remains fragile, especially under wireless communication and edge-deployment conditions. Third, although many studies emphasize real-time performance, they have not truly solved the problem of global synchronization instability caused by expired data, local distortion, and abnormal nodes [60,142,146].

Workflow orchestration determines the process through which agricultural multi-agent systems operate, resource scheduling determines who does what and to what extent under limited resources, and state synchronization determines whether the agricultural multi-agent system can always know how far execution has progressed and how the environment has changed. Together, these three aspects constitute the executional foundation of agricultural multi-agent systems. Only when this layer becomes stable can the previously discussed goal understanding, collaborative communication, and consensus mechanisms move beyond a purely theoretical structure and enter a stage of continuous, reliable, and scalable operation [56,60,142].

6. Future Outlook for Agricultural Agents

Although agricultural agents have demonstrated strong potential in environmental perception, task collaboration, intelligent decision-making, and equipment control, their transition from experimental validation to large-scale, stable, and sustainable application still faces a series of critical bottlenecks. As emerging paradigms such as multi-agent large models, cloud–edge–end collaboration, digital twins, and embodied intelligence continue to enter agricultural scenarios, the question is no longer simply whether the model itself is stronger, but whether these agents can obtain reliable data in complex environments, maintain cross-scenario adaptability, support stable multi-agent collaboration, and operate safely under software and hardware constraints. The treatment of challenges and future trends in the reference paper likewise suggests that what truly deserves emphasis is not methodological stacking itself but rather the common problems that determine whether a technology

can be effectively implemented. Therefore, the future development of agricultural agents can be discussed from five aspects: data extraction, model generalization, multi-agent collaboration, software–hardware collaboration, and security and trustworthiness.

6.1. Data Extraction Issues

The capabilities of agricultural agents are first constrained by their data foundation. Agricultural data sources are scattered, modalities are diverse, and spatiotemporal variations are strong. The same task often depends on multiple information sources such as remote sensing, unmanned aerial vehicles, field sensors, agricultural machinery logs, and textual knowledge. The current challenge is not simply that there is "not enough data," but rather that data collection, knowledge extraction, and sharing and circulation remain difficult, which directly limits the ability of agricultural agents to understand the real environment [10,147,148].

6.1.1. Challenges in Multimodal Agricultural Data Collection

Agricultural data collection is inherently dispersed and unstable. Satellites, UAVs, ground sensors, and agricultural machinery can jointly form a "sky–ground–human–machine" perception system, but their differences in sampling frequency, spatial resolution, and data format make unified representation difficult [47,149]. Data quality is also affected by occlusion, lighting changes, weather disturbances, and uneven target distribution. Therefore, future data collection for agricultural AI agents should emphasize cross-scale registration, dynamic quality control, and multisource collaborative perception, rather than simply increasing the number of devices.

6.1.2. Challenges in Agricultural Knowledge Extraction and Structuring

Agricultural agents must not only see the scene but also understand it. In practice, agricultural knowledge comes not only from expert experience, technical procedures, and scientific literature, but is also scattered across historical cases, farmers' practices, and operational records, thus exhibiting obvious heterogeneity and contextuality [53,147]. Although methods such as ontologies, knowledge graphs, and retrieval-augmented generation (RAG) provide important paths for structuring agricultural knowledge, agricultural knowledge itself has strong spatiotemporal dependence, and static rule libraries are difficult to use for covering complex scenarios. A more critical future direction is to promote agricultural knowledge from static database construction toward dynamic updating, so that knowledge extraction can truly support continuous reasoning and long-term decision-making [24,74].

6.1.3. Issues of Data Standardization and Sharing

Even when data can be collected and knowledge extracted, agricultural agents remain constrained by "data islands" if unified standards are lacking across devices, platforms, and participants. The main obstacles in current agricultural data systems include not only inconsistent formats but also mismatched semantic representations, insufficient interface compatibility, and unclear data-sharing boundaries [51,59,150]. Therefore, future agricultural data governance should not remain only at the communication-protocol or interface level but should gradually develop a collaborative framework integrating format standards, semantic standards, and sharing rules, thereby providing a more stable data foundation for cross-platform collaboration among agricultural agents.

6.2. Model Generalization Issues

Another core bottleneck of agricultural agents lies in the difficulty of maintaining stable model performance in open environments. Agricultural data are naturally characterized by strong spatiotemporal heterogeneity, and differences across regions, seasons, crop

stages, and even collection devices can all lead to distribution shifts. Therefore, model generalization is not a secondary issue but a key dividing line that determines whether agricultural agents can move from experimental conditions to real deployment [15,125,151].

6.2.1. Insufficient Cross-Scenario Task Generalization

Many agricultural models perform well in their training scenarios, yet their performance declines significantly when transferred to new plots, new seasons, or new equipment. The reason is that domain shift is widespread in agricultural scenarios, where differences in image style, background interference, and crop morphology all affect model stability [151]. In the future, this issue needs to be alleviated through domain generalization, multimodal fusion, and more robust representation learning so that models no longer rely excessively on local environmental features but can instead capture more stable agricultural semantics.

6.2.2. Insufficient Adaptability to Small-Sample and Long-Tail Tasks

Many critical agricultural tasks naturally suffer from insufficient samples, especially rare diseases, early stress symptoms, new varieties, and special disaster scenarios. The resulting long-tail distribution makes models more likely to learn common categories while struggling with truly critical low-frequency problems [152,153]. Transfer learning, meta-learning, and few-shot learning provide important ideas for addressing this issue, but their applicability in agriculture is still constrained by task differences and domain knowledge requirements. A more promising future direction is to combine few-shot learning with agricultural knowledge enhancement, rather than treating it merely as a temporary patch for insufficient data.

6.2.3. Insufficient Continuous Learning and Transfer Learning Capacity

The agricultural environment is continuously changing, while models are often trained once and applied over extended periods. Crop growth stages, pest and disease spectra, and environmental conditions evolve constantly, which means that agricultural agents do not encounter static tasks but operate within a dynamic target space [15,130]. Future agricultural agents will require not only task transfer capabilities but also continuous learning abilities, the capacity to incorporate new knowledge while minimizing forgetting of previously acquired information. Research on external memory, knowledge graphs, and retrieval-augmented generation (RAG) demonstrates promise, yet a fully mature mechanism for continuous learning in agricultural settings has yet to be established.

6.3. Multi-Agent Collaboration Issues

The real value of agricultural agents is often realized through the joint completion of complex tasks by multiple entities. However, multi-agent collaboration is not simply a matter of assembling several modules together; rather, it requires the collaborative framework to maintain a dynamic balance among efficiency, stability, and robustness. The heterogeneity, uncertainty, and resource-constrained nature of agricultural scenarios make collaboration one of the key challenges for future development [115,147].

6.3.1. Collaboration Efficiency and Communication Overhead

Increasing the number of agents can enable finer-grained division of labor, but it also increases the cost of state synchronization, task negotiation, and message transmission. Especially in wide-area farmlands and edge-network settings, excessive communication overhead often offsets the benefits brought by collaboration [19,59]. Therefore, the key to improving future collaborative efficiency lies not only in increasing communication speed but also in reducing ineffective communication through hierarchical collaboration structures and standardized protocols that compress interaction costs.

6.3.2. Stability Issues in Heterogeneous Agent Collaboration

Agricultural collaborative systems are usually composed of unmanned aerial vehicles, robots, sensors, edge devices, knowledge agents, and human participants. Different entities exhibit substantial differences in perception, control, and decision-making rhythms [105,115]. Therefore, the future focus should not be on making all entities "act in unison" but on constructing a collaborative framework that can tolerate differences, dynamically adjust roles, and maintain overall stability, so that heterogeneous entities can still form continuous and reliable collaborative chains in complex agricultural environments.

6.3.3. Fault Recovery and Robust Collaboration Issues

Faults in agricultural scenarios are not isolated incidents but common operating conditions. Sensor failures, equipment disconnections, network disruptions, and environmental changes can quickly interrupt the collaboration chain [56]. Future agricultural multi-agent systems should not merely design "optimal paths" but also incorporate failure recovery mechanisms to ensure task continuity, resource reconfigurability, and decision repairability, even when some agents fail. Robust collaboration should therefore evolve from device-level fault tolerance to system-level recovery.

6.4. Hardware–Software Coordination Issues

Agricultural AI agents must ultimately be deployed on field equipment and embedded into real task chains, making hardware–software coordination essential for practical operation. Under unstable networks, limited computing resources, and complex field environments, model performance alone cannot guarantee deployability. Future development will therefore depend on cloud–edge–end collaboration, equipment adaptation, and real-time computing optimization [60,125,154]. Figure 7 illustrates a cloud–edge–terminal–embodied collaborative architecture for agricultural AI agents.

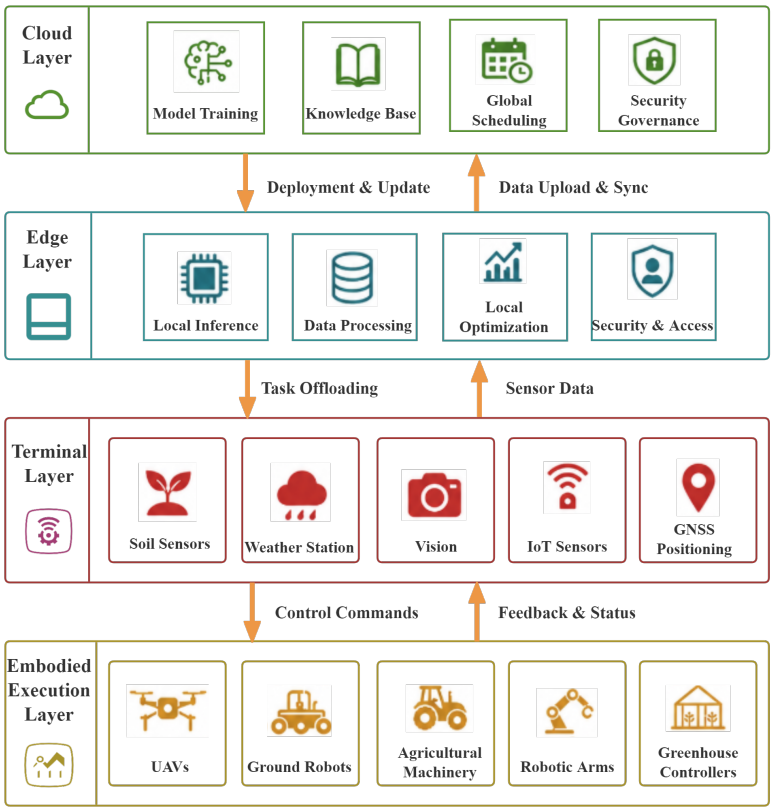


Figure 7. Cloud–edge–terminal–embodied architecture for agricultural agents.

6.4.1. Adaptation Issues of Cloud–Edge–End Coordination Architecture

It is difficult to reconcile with traditional centralized architectures the real-time and low-bandwidth requirements of agricultural tasks. As a result, cloud–edge–end coordination has become an important trend [60,154]. However, in agricultural scenarios, there are no universal answers to how tasks should be partitioned, when they should be offloaded, or how states should be synchronized. The key challenge for the future is not merely to "push some computation to the edge" but to establish an elastic coordination architecture that can dynamically adapt according to task complexity, network conditions, and resource availability.

6.4.2. Embodied Deployment Issues on Agricultural Equipment

Once agricultural agents are embedded in unmanned aerial vehicles, agricultural machinery, robotic arms, or greenhouse control systems, they must continuously interact with the physical environment. Agricultural settings are highly unstructured, and errors in perception, positioning, and control are considerably higher than in indoor or industrial assembly line scenarios [125,126]. Therefore, the challenge of embodied deployment is not merely "placing the model on the device" but ensuring that perception, planning, control, and mechanical systems form a continuous, reliable, and maintainable behavioral chain aligned with task objectives.

6.4.3. Issues of Real-Time Computing and Energy Constraints

A large number of devices in agricultural settings rely on batteries or low-power energy supply platforms, while agricultural tasks often impose strict real-time requirements. This makes the contradiction between real-time computing and energy constraints particularly prominent [60,154]. Future agricultural agents cannot rely solely on lightweight model compression; they also need to combine dynamic computing offloading, heterogeneous hardware collaboration, and the design of dedicated acceleration units in order to maintain stable and effective computing performance under limited energy budgets.

6.5. Security and Trust Issues

Whether agricultural agents can truly enter production practice in the future depends on whether they possess a sustainable foundation of security and trustworthiness. Risks in agricultural scenarios are not limited to model errors but also involve four interrelated levels: data, reasoning, systems, and governance. As multi-agent collaboration and automated execution continue to deepen, security and trust should no longer be viewed as external constraints but rather as inherent requirements in the design of agricultural AI agents [132,133,155].

6.5.1. Data Privacy and Data Property Protection Issues

Agricultural data are not only technical resources but also carry clear operational and economic value. Once farm records, environmental monitoring data, and operational data circulate among multiple entities, issues related to ownership, usage rights, and benefit distribution inevitably arise [132,155]. Techniques such as federated learning provide potential paths for privacy protection, but they do not automatically resolve questions of data ownership or governance boundaries. Future agricultural data security should therefore move beyond "leakage prevention" toward controlled sharing, legitimate use, and traceable rights.

6.5.2. Interpretability and Human–Machine Collaborative Supervision Issues

The recommendations generated by agricultural AI agents should not only be accurate but also understandable, contestable, and supervisable. Agricultural decision-making is

highly context-dependent. If the agricultural AI agent system cannot explain the basis of its judgments and their applicable boundaries, it will be difficult to gain the sustained trust of farmers and experts [155,156]. Therefore, interpretability in future systems should not remain at the level of result presentation but should instead support human–machine collaborative supervision, leaving room for human intervention and correction in high-risk and high-uncertainty stages.

6.5.3. Regulatory Compliance and Decision Traceability Issues

Beyond interpretability and human–machine collaborative supervision, agricultural agents must also satisfy emerging regulatory requirements related to data governance, transparency, accountability, and human oversight [132,155]. With the advancement of the European Union Artificial Intelligence Act (EU AI Act) and related policies on data access and governance, the Progress in Implementing the European Union Coordinated Plan on Artificial Intelligence further emphasizes the importance of trustworthy AI, data access, governance mechanisms, and cross-sector regulatory oversight. Therefore, agricultural agents should not only provide accurate decision outcomes but also ensure traceable and auditable decision-making processes, particularly in high-impact agricultural tasks such as pesticide application, irrigation scheduling, resource allocation, and equipment control.

To support regulatory compliance, agricultural agents need to record key process information throughout decision-making and execution, including data sources, reasoning chains, tool invocations, task workflows, and execution feedback. Mechanisms such as workflow logging, capability invocation tracking, human-in-the-loop supervision, and contract-driven planning can transform agent decisions into explainable, traceable, and verifiable execution processes [92]. Through these mechanisms, agricultural agents can better satisfy regulatory requirements for practical deployment while maintaining controllability, accountability, and trustworthiness.

6.5.4. Security Governance and Trust Evaluation Mechanism Issues

Agricultural AI agents usually operate in open systems composed of sensors, edge nodes, cloud platforms, and execution devices. Even if the model itself is correct, the overall system may still fail once communication links, sensing links, or execution links are attacked or tampered with [133]. Therefore, future agricultural AI agents need to establish a security governance framework spanning the entire chain of input, reasoning, execution, and feedback, while also introducing dynamic trust evaluation mechanisms to continuously assess the credibility of different data sources, agent roles, and output results. Only in this way can agricultural AI agents move from having strong performance to being reliably dependable over the long term.

6.6. Practical Deployment Barriers

Although agricultural agents have demonstrated strong technical potential in intelligent decision-making, autonomous collaboration, and adaptive agricultural management, their large-scale, real-world deployment still faces substantial practical barriers. Existing studies mainly emphasize model performance and collaborative intelligence, whereas actual agricultural environments are often constrained by limited budgets, unstable infrastructure, heterogeneous equipment, and insufficient technical support. In many agricultural regions, especially in small-scale farming systems, the main challenge is not whether advanced agricultural agents can be designed but whether they can be sustainably deployed, maintained, and practically adopted.

In addition, the lack of interoperability standards among different agricultural devices and platforms further limits cross-system collaboration, while issues related to data governance, responsibility attribution, and regulatory supervision remain insufficiently ad-

dressed. Adoption barriers among small-scale farmers are also particularly important since many intelligent agricultural systems still require considerable financial investment, digital literacy, and maintenance capability. Therefore, future agricultural agent research should move beyond purely algorithmic optimization and place greater emphasis on affordability, compatibility, maintainability, and accessibility under real agricultural conditions [125,130].

Moreover, many current agricultural intelligent agent systems are still evaluated under relatively constrained experimental environments, while their scalability, robustness, and long-term maintainability under real agricultural production conditions remain insufficiently validated [11,125]. This suggests that future agricultural intelligence research should place greater emphasis not only on improving algorithmic capability but also on ensuring sustainable deployment, operational reliability, and practical usability under real-world agricultural constraints [125,130].

7. Summary

Agricultural AI agents are gradually evolving from isolated task-oriented tools toward collaborative autonomous systems that integrate perception, reasoning, planning, execution, and feedback optimization in complex agricultural environments. This evolution reflects the transition of smart agriculture from single-function automation to system-level intelligence supported by multi-agent collaboration, domain knowledge, and continuous adaptation. In this process, agricultural AI agents are expected not only to improve the accuracy of individual tasks but also to enhance the coordination, flexibility, and resilience of the entire agricultural production system.

Despite recent progress in large language models, embodied intelligence, digital twins, and multi-agent collaboration, substantial challenges remain before agricultural AI agents can achieve large-scale practical deployment. Future research should therefore focus on several high-impact priorities, including robust multi-agent collaboration under dynamic agricultural conditions, scalable cloud–edge–end collaborative architectures, trustworthy and explainable decision-making, interoperability among heterogeneous agricultural devices, and low-cost deployment strategies suitable for real agricultural environments. In addition, improving accessibility and usability for small-scale farmers will be critical for promoting the sustainable adoption of agricultural AI agent systems.

Overall, the long-term value of agricultural AI agents will depend not only on continuous improvements in intelligence and autonomy but also on their affordability, reliability, maintainability, and practical integration into real-world agricultural production systems. Future agricultural AI agent research should therefore shift from merely pursuing algorithmic performance toward building deployable, trustworthy, and sustainable intelligent systems that can support resilient agricultural production and long-term agricultural modernization.

Author Contributions: Conceptualization, X.S. and Y.Z.; Methodology, X.S., Y.Z. and L.H.; Formal analysis, L.H., Q.W. and Z.Y.; Investigation, L.H., Q.W. and Z.Y.; Visualization, L.H.; Writing—original draft preparation, L.H., Q.W. and Z.Y.; Writing—review and editing, X.S., L.H., Y.Z., Q.W., Z.Y. and X.J.; Supervision, X.S., Y.Z. and X.J.; Project administration, X.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Jiangsu University, grant number JSU-JSJ-2025010.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: No new data were created or analyzed in this study.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Ge, C.; Zhang, G.; Wang, Y.; Shao, D.; Song, X.; Wang, Z. Research Status and Development Trends of Artificial Intelligence in Smart Agriculture. *Agriculture* **2025**, *15*, 2247. [CrossRef]
2. Zhu, Y.; Zhang, S.; Tang, S.; Gao, Q. Research Progress and Applications of Artificial Intelligence in Agricultural Equipment. *Agriculture* **2025**, *15*, 1703. [CrossRef]
3. Jiang, L.; Xu, B.; Husnain, N.; Wang, Q. Overview of Agricultural Machinery Automation Technology for Sustainable Agriculture. *Agronomy* **2025**, *15*, 1471. [CrossRef]
4. Duan, Y.; Han, W.; Guo, P.; Wei, X. YOLOv8-GDCI: Research on the Phytophthora Blight Detection Method of Different Parts of Chili Based on Improved YOLOv8 Model. *Agronomy* **2024**, *14*, 2734. [CrossRef]
5. Wu, J.; Li, C.; Pan, X.; Wang, X.; Zhao, X.; Gao, Y.; Yang, S.; Zhai, C. Model for Detecting Boom Height Based on an Ultrasonic Sensor for the Whole Growth Cycle of Wheat. *Agriculture* **2024**, *14*, 21. [CrossRef]
6. Wu, J.; Yang, S.; Gao, Y.; Pan, X.; Zou, W.; Wei, Y.; Zhai, C.; Chen, L. Optimization of a Boom Height Ultrasonic Detecting Model for the Whole Growth Cycle of Wheat: Affected by the Oscillation of the Three-Section Boom of the Sprayer. *Agriculture* **2024**, *14*, 1733. [CrossRef]
7. Yin, S.; Xi, Y.; Zhang, X.; Sun, C.; Mao, Q. Foundation Models in Agriculture: A Comprehensive Review. *Agriculture* **2025**, *15*, 847. [CrossRef]
8. Zhang, R.; Zhu, H.; Chang, Q.; Mao, Q. A Comprehensive Review of Digital Twins Technology in Agriculture. *Agriculture* **2025**, *15*, 903. [CrossRef]
9. Zhao, J.; Fan, S.; Zhang, B.; Wang, A.; Zhang, L.; Zhu, Q. Research Status and Development Trends of Deep Reinforcement Learning in the Intelligent Transformation of Agricultural Machinery. *Agriculture* **2025**, *15*, 1223. [CrossRef]
10. Murad, M.; Ahmed, M.; Din, N.U.; Shahid, M.F.; Siddiqui, S.; Byers, D.; Tanveer, M.H.; Voicu, R.C. Agentic AI Framework to Automate Traditional Farming for Smart Agriculture. *Agriengineering* **2026**, *8*, 8. [CrossRef]
11. Chen, H.; Hou, G.; Hua, C.; Wang, S.; Chen, Z.; Zhang, Y. Agricultural autonomous decision-making system "Fuxi Brain" Based on generative large model fusion internet of things. *Comput. Electron. Agric.* **2026**, *244*, 111454. [CrossRef]
12. Qin, S.; Zhang, S.; Zhong, W.; He, Z. Control Algorithms for Intelligent Agriculture: Applications, Challenges, and Future Directions. *Processes* **2025**, *13*, 3061. [CrossRef]
13. Wu, Y.; Chen, L.; Yang, N.; Sun, Z. Research Progress of Deep Learning-Based Artificial Intelligence Technology in Pest and Disease Detection and Control. *Agriculture* **2025**, *15*, 2077. [CrossRef]
14. Wang, Y.; Zhang, Z.; Jia, W.; Ou, M.; Dong, X.; Dai, S. A Review of Environmental Sensing Technologies for Targeted Spraying in Orchards. *Horticulturae* **2025**, *11*, 551. [CrossRef]
15. Zhao, Y.; Liang, J.; Chen, B.; Deng, X.; Zhang, Y.; Xiong, Z.; Pan, M.; Meng, X. Applications Research Progress and Prospects of Multi-Agent Large Language Models in Agricultural. *Smart Agric.* **2025**, *7*, 37–51. [CrossRef]
16. Zhang, Y.; Zhang, B.; Shen, C.; Liu, H.; Huang, J.; Tian, K.; Tang, Z. Review of the field environmental sensing methods based on multi-sensor information fusion technology. *Int. J. Agric. Biol. Eng.* **2024**, *17*, 1–13. [CrossRef]
17. Wan, L.; Li, H.; Li, C.; Wang, A.; Yang, Y.; Wang, P. Hyperspectral Sensing of Plant Diseases: Principle and Methods. *Agronomy* **2022**, *12*, 1451. [CrossRef]
18. Goodrich, P.; Betancourt, O.; Arias, A.C.; Zohdi, T. Placement and drone flight path mapping of agricultural soil sensors using machine learning. *Comput. Electron. Agric.* **2023**, *205*, 107591. [CrossRef]
19. Betalo, M.L.; Leng, S.; Abishu, H.N.; Dharejo, F.A.; Seid, A.M.; Erbad, A.; Naqvi, R.A.; Zhou, L.; Guizani, M. Multi-Agent Deep Reinforcement Learning-Based Task Scheduling and Resource Sharing for O-RAN-Empowered Multi-UAV-Assisted Wireless Sensor Networks. *IEEE Trans. Veh. Technol.* **2024**, *73*, 9247–9261. [CrossRef]
20. Zou, Y.; Quan, L. Resource management and scheduling policy based on grid for AIoT. *Mod. Phys. Lett. B* **2017**, *31*, 1740066. [CrossRef]
21. Gonzalez-Briones, A.; Mezquita, Y.; Castellanos-Garzon, J.A.; Prieto, J.; Corchado, J.M. Intelligent multi-agent system for water reduction in automotive irrigation processes. *Procedia Comput.* **2019**, *151*, 971–976. [CrossRef]
22. Agyeman, B.T.; Naouri, M.; Appels, W.M.; Liu, J.; Shah, S.L. Learning-based multi-agent MPC for irrigation scheduling. *Control Eng. Pract.* **2024**, *147*, 105908. [CrossRef]
23. Ge, W.; Zhou, J.; Zheng, P.; Yuan, L.; Rottok, L.T. A recommendation model of rice fertilization using knowledge graph and case-based reasoning. *Comput. Electron. Agric.* **2024**, *219*, 108751. [CrossRef]
24. Lin, Y.; Li, D.; Peng, P.; Liang, J.; Ding, F.; Jin, X.; Zeng, Z. A reasoning method for rice fertilization strategy based on spatiotemporal knowledge graph. *Trans. GIS* **2024**, *28*, 902–924. [CrossRef]
25. Li, K.; Shi, J.; Hu, C.; Xue, W. The Intelligentization Process of Agricultural Greenhouse: A Review of Control Strategies and Modeling Techniques. *Agriculture* **2025**, *15*, 2135. [CrossRef]
26. Zhang, S. Chat Demeter: A multi-agent system for plant disease diagnosis integrating CNN-transformer models. *Front. Plant Sci.* **2026**, *16*, 1695227. [CrossRef]

27. Anand, A.; Shen, J.; Cremers, A.B.; Jacobs, M. Algorithms in the orchard: An embedding-based expert answering system for apple rust. *Smart Agric. Technol.* **2025**, *12*, 101069. [CrossRef]
28. Owiti, T.; Kipkebut, A. Enhancing AI-Driven Farming Advisory in Kenya with Efficient RAG Agents via Quantized Fine-Tuned Language Models. In *Proceedings of the Sixth Workshop on African Natural Language Processing, AFRICANLP 2025*; Association for Computational Linguistics: Stroudsburg, PA, USA, 2025; pp. 24–30. [CrossRef]
29. Jayarathna, H.M.H.R.; Hettige, B. AgriCom: A Communication Platform for Agriculture Sector. In *Proceedings of the 2013 8th IEEE International Conference on Industrial and Information Systems (ICIIS)*; IEEE: New York, NY, USA, 2013; pp. 439–444. [CrossRef]
30. Skobelev, P.; Mayorov, I.; Simonova, E.; Laryukhin, V.; Yalovenko, O. Vertical and Horizontal Negotiations of Multi-Agent Planning Services in a Multi-Service Platform for Crop Managing. In *Proceedings of the 2019 XXI International Conference Complex Systems: Control and Modeling Problems (CSCMP)*; IEEE: New York, NY, USA, 2019; pp. 78–83. [CrossRef]
31. Guo, H.; Pan, Q.; Sang, H.; Miao, Z.; Zhang, W. AHLLNS: An Automated Algorithm for Multi-Objective Heterogeneous Agricultural Robot Operation Scheduling Problems. *IEEE Trans. Autom. Sci. Eng.* **2026**, *23*, 5096–5109. [CrossRef]
32. Jo, Y.; Son, H.I. A Path Planning and Coordination Algorithm for Heterogeneous Tasks of Multi-UGV in Smart Farm: Work in Progress. In *Proceedings of the 2022 22nd International Conference on Control, Automation and Systems (ICCAS 2022)*; IEEE: New York, NY, USA, 2022; pp. 1387–1390. [CrossRef]
33. Wu, H.; Wang, X.; Chen, X.; Zhang, Y.; Zhang, Y. Review on Key Technologies for Autonomous Navigation in Field Agricultural Machinery. *Agriculture* **2025**, *15*, 1297. [CrossRef]
34. Jin, Y.; Liu, J.; Xu, Z.; Yuan, S.; Li, P.; Wang, J. Development status and trend of agricultural robot technology. *Int. J. Agric. Biol. Eng.* **2021**, *14*, 1–19. [CrossRef]
35. Chen, Z.; Yin, J.; Farhan, S.M.; Liu, L.; Zhang, D.; Zhou, M.; Cheng, J. A comprehensive review of obstacle avoidance for autonomous agricultural machinery in multi-ope rational environment. *Artif. Intell. Agric.* **2026**, *16*, 139–163. [CrossRef]
36. Wu, M.; Liu, S.; Li, Z.; Ou, M.; Dai, S.; Dong, X.; Wang, X.; Jiang, L.; Jia, W. A Review of Intelligent Orchard Sprayer Technologies: Perception, Control, and System Integration. *Horticulturae* **2025**, *11*, 668. [CrossRef]
37. Jia, H.; Chen, W.; Su, Z.; Sun, Y.; Qian, Z.; Huang, L. AI-Driven Cooperative Control for Autonomous Tractors and Implements: A Comprehensive Review. *Agriengineering* **2025**, *7*, 394. [CrossRef]
38. Salah, K.; Chen, X.; Neshatian, K.; Pretty, C. A Hybrid Control Multi-Agent Cooperative System for Autonomous Bin Transport during Apple Harvest. In *Proceedings of the 2018 13th IEEE Conference on Industrial Electronics and Applications (ICIEA 2018)*; IEEE: New York, NY, USA, 2018; pp. 644–649. [CrossRef]
39. Ma, S.; Li, R.; Cong, S.; Nie, X.; Khan, R.A.I.; Yu, C.; Zhou, W.; Hu, H.; Fang, H.; Shang, H. Autonomous Weeding Robots: Coordinating Locomotion and Manipulation via Long Short-Term Memory-Proximal Policy Optimization Reinforcement Learning. *IEEE Trans. Autom. Sci. Eng.* **2026**, *23*, 6473–6486. [CrossRef]
40. Jo, Y.; Son, H.I. CBS-HT: Prioritized Safe Interval Path-Planning Algorithm for Heterogeneous Agricultural Robot Team. *IEEE Access* **2025**, *13*, 146630–146648. [CrossRef]
41. Manasherov, O.; Degani, A. Multi-Agent target allocation and safe trajectory planning for artificial pollination tasks. *Smart Agric. Technol.* **2024**, *8*, 100461. [CrossRef]
42. Li, T.; Xie, F.; Qiu, Q.; Feng, Q. Multi-Arm Robot Task Planning for Fruit Harvesting Using Multi-Agent Reinforcement Learning. In *Proceedings of the 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS*; IEEE: New York, NY, USA, 2023; pp. 4176–4183. [CrossRef]
43. Ma, Z.; Wang, X.; Chen, X.; Hu, B.; Li, J. Advances in Crop Row Detection for Agricultural Robots: Methods, Performance Indicators, and Scene Adaptability. *Agriculture* **2025**, *15*, 2151. [CrossRef]
44. Goral, P.; Pawlowski, P.; Piniarski, K.; Dabrowski, A. Multi-Agent Vision System for Supporting Autonomous Orchard Spraying. *Electronics* **2024**, *13*, 494. [CrossRef]
45. Ankit, K.; Kolathaya, S.N.Y.; Ghose, D. Multi-Agent Collaborative Framework for Automated Agriculture. In *Proceedings of the 2021 15th International Conference on Advanced Computing and Applications (ACOMP 2021)*; IEEE: New York, NY, USA, 2021; pp. 30–37. [CrossRef]
46. Li, J.; Zhang, W.; Ren, J.; Yu, W.; Wang, G.; Ding, P.; Wang, J.; Zhang, X. A Multi-Area Task Path-Planning Algorithm for Agricultural Drones Based on Improved Double Deep Q-Learning Net. *Agriculture* **2024**, *14*, 1294. [CrossRef]
47. Morchid, A.; El Alami, R.; Raezah, A.A.; Sabbar, Y. Applications of internet of things (IoT) and sensors technology to increase food security and agricultural Sustainability: Benefits and challenges. *Ain Shams Eng. J.* **2024**, *15*, 102509. [CrossRef]
48. Zhao, T.; Zhang, P.; Hou, H. The Agricultural Irrigation District Information System based on Multi-Agent and GSM. *Appl. Mech. Mater.* **2013**, *433*, 1853–1856. [CrossRef]
49. Ikidid, A.; El Fazziki, A.; Sadgal, M. Smart Collective Irrigation: Agent and Internet of Things based system. In *Proceedings of the 13th International Conference on Management of Digital Ecosystems, MEDES 2021*; Association for Computing Machinery: New York, NY, USA, 2020; pp. 100–106. [CrossRef]

50. Kalyani, Y.; Collier, R. The Role of Multi-Agents in Digital Twin Implementation: Short Survey. *ACM Comput. Surv.* **2025**, *57*, 72. [CrossRef]
51. Kalyani, Y.; Collier, R. Towards a New Architecture: Multi-agent Based Cloud-Fog-Edge Computing and Digital Twin for Smart Agriculture. In *Proceedings of the Intelligent Distributed Computing XV, IDC 2022*; Springer: Cham Switzerland, 2023; Volume 1089, pp. 111–117. [CrossRef]
52. Kalyani, Y.; Vorster, L.; Whetton, R.; Collier, R. Application Scenarios of Digital Twins for Smart Crop Farming through Cloud-Fog-Edge Infrastructure. *Future Internet* **2024**, *16*, 100. [CrossRef]
53. Skobelev, P.O.; Simonova, E.V.; Smirnov, S.V.; Budaev, D.S.; Voshchuk, G.Y.; Morokov, A.L. Development of a Knowledge Base in the "Smart Farming" System for Agricultural Enterprise Management. *Procedia Comput. Sci.* **2019**, *150*, 154–161. [CrossRef]
54. Ramanathan, G.; Vachtsevanou, D.; Garcia, K.; Lemee, J.; Burattini, S.; Bekta, K.; Mayer, S. Semantic Knowledge for Autonomous Smart Farming. *IFAC-Pap.* **2022**, *55*, 217–222. [CrossRef]
55. Kahneman, D. *Thinking, Fast and Slow*; Farrar, Straus and Giroux: New York, NY, USA, 2011.
56. Yang, B.; Zhang, Y.; Chen, Y.; Feng, L.; Xu, X.; Aierken, N.; Li, S. AgriAgent: Contract-Driven Planning and Capability-Aware Tool Orchestration in Real-World Agriculture. *arXiv* **2026**, arXiv:2601.08308. [CrossRef]
57. Harman, H.; Sklar, E.I. Challenges for Multi-Agent Based Agricultural Workforce Management. In *Proceedings of the Multi-Agent-Based Simulation XXIII, MABS 2022*; Springer: Cham, Switzerland, 2023; Volume 13743, pp. 121–133. [CrossRef]
58. Harman, H.; Sklar, E.I. Multi-agent task allocation for harvest management. *Front. Robot. AI* **2022**, *9*, 864745. [CrossRef]
59. Guo, W.; Li, W.; Zhong, Y.; Lodewijks, G.; Shen, W. Agent-based Negotiation Framework for Agricultural Supply Chain Supported by Third Party Logistics. In *Proceedings of the 2016 IEEE 20th International Conference on Computer Supported Cooperative Work in Design (CSCWD)*; IEEE: New York, NY, USA, 2016; pp. 584–589. [CrossRef]
60. Yu, P.; Teng, F.; Zhu, W.; Shen, C.; Chen, Z.; Song, J. Cloud–edge–device collaborative computing in smart agriculture: Architectures, applications, and future perspectives. *Front. Plant Sci.* **2025**, *16*, 1668545. [CrossRef]
61. Zhu, A.; Zeng, Z.; Guo, S.; Lu, H.; Ma, M.; Zhou, Z. Game-theoretic robotic offloading via multi-agent learning for agricultural applications in heterogeneous networks. *Comput. Electron. Agric.* **2023**, *211*, 108017. [CrossRef]
62. Islam, A.; Ghose, M. GSAgri: Green and Secure Agriculture through efficient task offloading and scheduling under IoT-enabled energy-harvesting multi-access edge computing framework. *Expert Syst. Appl.* **2025**, *284*, 127814. [CrossRef]
63. Tan, H.; Zhang, Q.; Li, M.; Liu, X.; Hu, L. Dynamic resource allocation in smart agricultural IoT using reinforcement learning and 6g edge computing. *EURASIP J. Wirel. Commun. Netw.* **2026**, *2026*, 23. [CrossRef]
64. Jogeshwar, B.K.; Sevil, H.E.; Haghshenas-Jaryani, M. Multi-Agent Coverage Planning for Agricultural Soil and Crop Monitoring. In *Proceedings of the ASME 2025 International Mechanical Engineering Congress and Exposition, IMECE2025*; ASME: New York, NY, USA, 2025; Volume 5. [CrossRef]
65. Salazar, R.; Carlos Rangel, J.; Pinzon, C.; Rodriguez, A. Irrigation System through Intelligent Agents Implemented with Arduino Technology. *ADCAIJ-Adv. Distrib. Comput. Artif. Intell. J.* **2013**, *2*, 29–36. [CrossRef]
66. Liu, Y.; Yan, H.; Zhang, C.; Zhang, J.; Wang, G.; Zhang, D.; Bao, R.; Han, Y. Application of Irrigation Decision-Making Methods for Smart Irrigation Systems: A Review. *Irrig. Drain.* **2026**. [CrossRef]
67. Seralathan, P.; Edward, A.S. Reinforcement learning based dynamic vegetation index formulation for rice crop stress detection using satellite and mobile imagery. *Sci. Rep.* **2025**, *16*, 3447. [CrossRef]
68. Bahri, O.; Mourhir, A.; Papageorgiou, E.I. Integrating fuzzy cognitive maps and multi-agent systems for sustainable agriculture. *EURO-Mediterr. J. Environ. Integr.* **2020**, *5*, 7. [CrossRef]
69. Xu, J.; Liu, H.; Shen, Y. Image and Point Cloud-Based Neural Network Models and Applications in Agricultural Nursery Plant Protection Tasks. *Agronomy* **2025**, *15*, 2147. [CrossRef]
70. Ghazal, S.; Munir, A.; Qureshi, W.S. Computer vision in smart agriculture and precision farming: Techniques and applications. *Artif. Intell. Agric.* **2024**, *13*, 64–83. [CrossRef]
71. Weyler, J.; Laebe, T.; Magistri, F.; Behley, J.; Stachniss, C. Towards Domain Generalization in Crop and Weed Segmentation for Precision Farming Robots. *IEEE Robot. Autom. Lett.* **2023**, *8*, 3310–3317. [CrossRef]
72. Angarano, S.; Martini, M.; Navone, A.; Chiaberge, M. Domain Generalization for Crop Segmentation with Standardized Ensemble Knowledge Distillation. In *Proceedings of the 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, CVPRW*; IEEE: New York, NY, USA, 2024; pp. 5450–5459. [CrossRef]
73. Ning, Y.; Junjie, Y.; Aiying, W.; Jian, T.; Rongbiao, Z.; Liangliang, X.; Fangyu, S.; Kwabena, O.P. A rapid rice blast detection and identification method based on crop disease spores' diffraction fingerprint texture. *J. Sci. Food Agric.* **2020**, *100*, 3608–3621. [CrossRef]
74. Ali, N.; Ullah, M.; Mohammed, A.; Bais, A.; Berraies, S.; Ruan, Y.; Cuthbert, R.D.; Sangha, J.S. Advancing Fusarium Head Blight Detection in Wheat Crop: A Review and Future Directions to Sustainable Agriculture. *IEEE Trans. Consum. Electron.* **2025**, *71*, 7473–7491. [CrossRef]

75. Bhujade, V.G.; Shrawne, S.C.; Sambhe, V.K. Implementation and Performance Evaluation of Deep Learning Models for Disease Classification and Severity Estimation of Coffee Leaves. In *Proceedings of the Advanced Network Technologies and Intelligent Computing, ANTIC 2023, PT III*; Springer: Cham, Switzerland, 2024; Volume 2092, pp. 3–19. [CrossRef]
76. Mostafa, S.A.; Hazeem, A.A.; Khaleefahand, S.H.; Mustapha, A.; Darman, R. A Collaborative Multi-agent System for Oil Palm Pests and Diseases Global Situation Awareness. In *Proceedings of the Future Technologies Conference (FTC) 2018, VOL 1*; Springer: Cham, Switzerland, 2019; Volume 880, pp. 763–775. [CrossRef]
77. Zhou, J.W.; Zou, X.M.; Song, S.H.; Chen, G.H. Quantum Dots Applied to Methodology on Detection of Pesticide and Veterinary Drug Residues. *J. Agric. Food Chem.* **2018**, *66*, 1307–1319. [CrossRef]
78. Wang, P.; Li, H.; Hassan, M.M.; Guo, Z.; Zhang, Z.Z.; Chen, Q. Fabricating an Acetylcholinesterase Modulated UCNPs-Cu2+ Fluorescence Biosensor for Ultrasensitive Detection of Organophosphorus Pesticides-Diazinon in Food. *J. Agric. Food Chem.* **2019**, *67*, 4071–4079. [CrossRef] [PubMed]
79. Chen, Z.; Sun, Y.; Shi, J.; Zhang, W.; Zhang, X.; Huang, X.; Zou, X.; Li, Z.; Wei, R. Facile synthesis of Au@Ag core-shell nanorod with bimetallic synergistic effect for SERS detection of thiabendazole in fruit juice. *Food Chem.* **2022**, *370*, 131276. [CrossRef]
80. Wang, L.; Haruna, S.A.; Ahmad, W.; Wu, J.; Chen, Q.; Ouyang, Q. Tunable multiplexed fluorescence biosensing platform for simultaneous and selective detection of paraquat and carbendazim pesticides. *Food Chem.* **2022**, *388*, 132950. [CrossRef] [PubMed]
81. Aheto, J.H.; Huang, X.; Wang, C.; Tian, X.; Yi, R.; Yuena, W. Fabrication and evaluation of chitosan modified filter paper for chlorpyrifos detection in wheat by surface-enhanced Raman spectroscopy. *J. Sci. Food Agric.* **2022**, *102*, 7323–7330. [CrossRef]
82. Guo, Z.; Wu, X.; Jayan, H.; Yin, L.; Xue, S.; El-Seedi, H.R.; Zou, X. Recent developments and applications of surface enhanced Raman scattering spectroscopy in safety detection of fruits and vegetables. *Food Chem.* **2024**, *434*, 137469. [CrossRef] [PubMed]
83. Marimuthu, M.; Xu, K.; Song, W.; Chen, Q.; Wen, H. Safeguarding food safety: Nanomaterials-based fluorescent sensors for pesticide tracing. *Food Chem.* **2025**, *463*, 141288. [CrossRef]
84. Awais, M.; Li, W.; Hussain, S.; Cheema, M.J.M.; Li, W.; Song, R.; Liu, C. Comparative Evaluation of Land Surface Temperature Images from Unmanned Aerial Vehicle and Satellite Observation for Agricultural Areas Using In Situ Data. *Agriculture* **2022**, *12*, 184. [CrossRef]
85. Oudendag, D.; Hoogendoorn, M.; Jongeneel, R. Agent-Based Modeling of Farming Behavior: A Case Study for Milk Quota Abolishment. In *Proceedings of the Modern Advances in Applied Intelligence, IEA/AIE 2014, PT I*; Springer: Cham, Switzerland, 2014; Volume 8481, pp. 11–20. [CrossRef]
86. Ito, J. Impediments to efficient land reallocation in agriculture: Multi-agent simulation model of transaction costs and farm retirement. *Land Degrad. Dev.* **2024**, *35*, 1553–1566. [CrossRef]
87. Berger, T. Agent-based spatial models applied to agriculture: A simulation tool for technology diffusion, resource use changes and policy analysis. *Agric. Econ.* **2001**, *25*, 245–260. [CrossRef]
88. Imane, B.; El Miloud, J.; Abdelmajid, B.; Mohammed, T.A. Agent Mining Framework for Analyzing Moroccan Olive Oil Datasets. *Int. J. Adv. Comput. Sci. Appl.* **2020**, *11*, 638–646. [CrossRef]
89. Singh, A.; Sharma, A. A Framework for Semantics and Agent Based Personalized Information Retrieval in Agriculture. In *Proceedings of the 2015 2nd International Conference on Computing for Sustainable Global Development (INDIACOM)*; IEEE: New York, NY, USA, 2015; pp. 929–931.
90. Nguyen, V.G.N.; Drogoul, A.; Huynh, H.X. Toward an Agent-based Multi-scale Recommendation System for Brown Plant Hopper Control. In *Proceedings of the 2012 Sixth Uksim/Amss European Symposium on Computer Modelling and Simulation (EMS)*; IEEE: New York, NY, USA, 2012; pp. 9–14. [CrossRef]
91. Akbar, N.A.; Dembani, R.; Lenzitti, B.; Tegolo, D. RAG-Driven Memory Architectures in Conversational LLMs-A Literature Review with Insights Into Emerging Agriculture Data Sharing. *IEEE Access* **2025**, *13*, 123855–123880. [CrossRef]
92. Song, Y.; Lv, C.; Liu, J. Quality and safety traceability system of agricultural products based on Multi-agent. *J. Intell. Fuzzy Syst.* **2018**, *35*, 2731–2740. [CrossRef]
93. Su, X.; Duan, S.; Guo, S.; Liu, H. Evolutionary Games in the Agricultural Product Quality and Safety Information System: A Multiagent Simulation Approach. *Complexity* **2018**, 7684185. [CrossRef]
94. Ding, C.; Wang, L.; Chen, X.; Yang, H.; Huang, L.; Song, X. A Blockchain-Based Wide-Area Agricultural Machinery Resource Scheduling System. *Appl. Eng. Agric.* **2023**, *39*, 1–12. [CrossRef]
95. Zhang, P.; Li, N.; Han, H. An Evolutionary Game Study of Multi-Agent Collaborative Disaster Relief Mechanisms for Agricultural Natural Disasters in China. *Sustainability* **2025**, *17*, 7194. [CrossRef]
96. Jimenez, A.F.; Cardenas, P.F.; Canales, A.; Jimenez, F.; Portacio, A. A survey on intelligent agents and multi-agents for irrigation scheduling. *Comput. Electron. Agric.* **2020**, *176*, 105474. [CrossRef]
97. Ma, Y.W.; Shi, J.Q.; Chen, J.L.; Hsu, C.C.; Chuang, C.H. Integration Agricultural Knowledge and Internet of Things for Multi-Agent Deficit Irrigation Control. In *Proceedings of the 2019 21st International Conference on Advanced Communication Technology (ICACT): ICT for 4th Industrial Revolution*; IEEE: New York, NY, USA, 2019; pp. 299–304. [CrossRef]

98. Khorshidi, M.S.; Izady, A.; Nikoo, M.R.; Al-Maktoumi, A.; Chen, M.; Gandomi, A.H. An Agent-based Framework for Transition from Traditional to Advanced Water Supply Systems in Arid Regions. *Water Resour. Manag.* **2024**, *38*, 2565–2579. [CrossRef]
99. Dong, J.; Dong, H.; Sun, S.; Liu, J.; Zhang, J.; Jia, C.; Wang, Y. Agricultural water-carbon footprint regulation based on DNDC model and multi-agent model fusion gradient boosting algorithm. *Eur. J. Agron.* **2026**, *172*, 127867. [CrossRef]
100. Zhai, Z.; Martinez Ortega, J.F.; Lucas Martinez, N.; Rodriguez-Molina, J. A Mission Planning Approach for Precision Farming Systems Based on Multi-Objective Optimization. *Sensors* **2018**, *18*, 1795. [CrossRef]
101. Nagarsheth, S.; Agbossou, K.; Henao, N.; Bendouma, M. The Advancements in Agricultural Greenhouse Technologies: An Energy Management Perspective. *Sustainability* **2025**, *17*, 3407. [CrossRef]
102. Hindi, I.; Alsharkawi, A.; Al-Ajlouni, M.; Qarallah, B. Enhancing autonomous agriculture control systems in greenhouses for sustainable resource usage using deep learning techniques. *PLoS ONE* **2026**, *21*, e0344946. [CrossRef] [PubMed]
103. Taha, M.F.; Mao, H.; Zhang, Z.; Elmasry, G.; Awad, M.A.; Abdalla, A.; Mousa, S.; Elwakeel, A.E.; Elsherbiny, O. Emerging Technologies for Precision Crop Management Towards Agriculture 5.0: A Comprehensive Overview. *Agriculture* **2025**, *15*, 582. [CrossRef]
104. Xiang, R.; Ma, P.; Gong, L.; Tian, M.; Li, R.; Fei, H.; Yi, H. ASS-DQN: A deep reinforcement learning approach for path planning in multi-machine cooperative operations. *Smart Agric. Technol.* **2026**, *13*, 101887. [CrossRef]
105. Lee, H.; Bae, J.; Patil, A.; Park, M.; Nguyen, V. Heuristic Approaches for Coordinating Collaborative Heterogeneous Robotic Systems in Harvesting Automation with Size Constraints. *Sensors* **2025**, *25*, 6443. [CrossRef] [PubMed]
106. Gallou, J.; Lippi, M.; Palmieri, J.; Gasparri, A.; Marino, A. A Human-Centered Task Allocation and Scheduling Framework for Multi-Human-Multi-Robot Collaboration in Precision Agriculture Settings. *IEEE Trans. Autom. Sci. Eng.* **2025**, *22*, 20126–20145. [CrossRef]
107. Jo, Y.; Son, H.I. Field Evaluation of a Prioritized Path-Planning Algorithm for Heterogeneous Agricultural Tasks of Multi-UGVs. In *Proceedings of the 2024 IEEE International Conference on Robotics and Automation (ICRA 2024)*; IEEE: New York, NY, USA, 2024; pp. 11891–11897. [CrossRef]
108. Tang, R.; Tang, J.; Talip, M.S.A.; Aridas, N.K.; Xu, X. Enhanced multi agent coordination algorithm for drone swarm patrolling in durian orchards. *Sci. Rep.* **2025**, *15*, 9139. [CrossRef]
109. Ge, G.; Sun, M.; Xue, Y.; Pavlova, S. Transformer-Based Soft Actor-Critic for UAV Path Planning in Precision Agriculture IoT Networks. *Sensors* **2025**, *25*, 7463. [CrossRef]
110. Yu, Y.; Li, Z.; Dai, B.; Pan, J.; Xu, L. High-Precision Mapping and Real-Time Localization for Agricultural Machinery Sheds and Farm Access Roads Environments. *Agriculture* **2025**, *15*, 2248. [CrossRef]
111. Nivetha, R.; Sriharipriya, K.C.; Balusamy, B. Self-Supervised Learning Graphical Neural Network Driven Prediction Model for Path-Planning and Navigation in Smart Sustainable Agriculture. *IEEE Access* **2025**, *13*, 151235–151257. [CrossRef]
112. Ahmed, S.; Qiu, B.; Ahmad, F.; Kong, C.W.; Xin, H. A State-of-the-Art Analysis of Obstacle Avoidance Methods from the Perspective of an Agricultural Sprayer UAV's Operation Scenario. *Agronomy* **2021**, *11*, 1069. [CrossRef]
113. Yao, X.; Cui, Y.; Xu, G.; Fan, S.; Wei, H. Research on Intelligent Management of Collaborative Relation in Agricultural Products Logistics. In *Proceedings of the ICICTA: 2009 Second International Conference on Intelligent Computation Technology and Automation, VOL III, PROCEEDINGS*; IEEE: New York, NY, USA, 2009; pp. 963–966. [CrossRef]
114. Goel, A.; Zobel, C.; Jones, E. A multi-agent system for supporting the electronic contracting of food grains. *Comput. Electron. Agric.* **2005**, *48*, 123–137. [CrossRef]
115. Qian, Y.C.; Miao, Z.H.; Zhou, J.; Zhu, X.J. Leader-follower consensus of nonlinear agricultural multiagents using distributed adaptive protocols. *Adv. Manuf.* **2025**, *13*, 901–910. [CrossRef]
116. Sang, J.; Ma, D.; Xie, X.; Hu, X. Group-aggregation of Hierarchical Containment Control for Homogeneous Multi-agent Systems in Precision Agriculture. *Int. J. Control Autom. Syst.* **2024**, *22*, 1400–1408. [CrossRef]
117. Munteanu, S.; Sudacevschi, V.; Ababii, V.; Carbune, V.; Borozan, O.; Alexei, V. Self-Organizing Multi-Agent Collaborative Decision-Making System. In *Proceedings of the 2025 International Conference on Electromechanical and Energy Systems, SIELMEN*; IEEE: New York, NY, USA, 2025; pp. 314–319. [CrossRef]
118. Wang, B.; Du, X.; Wang, Y.; Mao, H. Multi-machine collaboration realization conditions and precise and efficient production mode of intelligent agricultural machinery. *Int. J. Agric. Biol. Eng.* **2024**, *17*, 27–36. [CrossRef]
119. Khan, Z.; Shen, Y.; Liu, H. ObjectDetection in Agriculture: A Comprehensive Review of Methods, Applications, Challenges, and Future Directions. *Agriculture* **2025**, *15*, 1351. [CrossRef]
120. Gao, X.; Gao, J.; Qureshi, W. Applications, Trends, and Challenges of Precision Weed Control Technologies Based on Deep Learning and Machine Vision. *Agronomy* **2025**, *15*, 1954. [CrossRef]
121. Zhu, X.; Chikangaise, P.; Shi, W.; Chen, W.H.; Yuan, S. Review of Intelligent Sprinkler Irrigation Technologies for Remote Autonomous System. *Int. J. Agric. Biol. Eng.* **2018**, *11*, 23–30. [CrossRef]
122. Jiang, Q.; Shen, Y.; Liu, H.; Khan, Z.; Sun, H.; Huang, Y. A Hybrid Path Planning Algorithm for Orchard Robots Based on an Improved D* Lite Algorithm. *Agriculture* **2025**, *15*, 1698. [CrossRef]

123. Xie, B.; Jin, Y.; Faheem, M. Research Progress of Autonomous Navigation Technology for Multi-Agricultural Scenes. *Comput. Electron. Agric.* **2023**, *211*, 107963. [CrossRef]
124. Liu, J.; Abbas, I.; Noor, R. Development of Deep Learning-Based Variable Rate Agrochemical Spraying System for Targeted Weeds Control in Strawberry Crop. *Agronomy* **2021**, *11*, 1480. [CrossRef]
125. Li, Y.; Wang, J.; Yuan, Z.; Zhang, H. Agent Technology for Agricultural Intelligence: Methodological Framework and Applications. *Electronics* **2026**, *15*, 1547. [CrossRef]
126. Ju, C.; Kim, J.; Seol, J.; Son, H.I. A review on multirobot systems in agriculture. *Comput. Electron. Agric.* **2022**, *202*, 107336. [CrossRef]
127. Park, Y.; Son, H.I. Visual Scene Understanding-Based Task Planning for an Efficient Multipurpose Agricultural Robot System. *IEEE Robot. Autom. Lett.* **2025**, *10*, 13241–13248. [CrossRef]
128. Zuzuárregui, M.A.; Toslak, M.M.; Carpin, S. One For All: LLM-based Heterogeneous Mission Planning in Precision Agriculture. *IFAC-PapersOnLine* **2025**, *59*, 344–349. [CrossRef]
129. Gassen, E.; Meckel, D.; Oladimeji, O.; Berns, K. Ontologies for Autonomous Agricultural Robots and Implements. In *Proceedings of the 2025 5th International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)*; IEEE: New York, NY, USA, 2025; pp. 1–5. [CrossRef]
130. Thudumu, S.; Fisher, J. OpenAg: Democratizing Agricultural Intelligence. *arXiv* **2025**, arXiv:2506.04571. [CrossRef]
131. Li, L.-X.; Sang, H.-Y.; Zhang, D. An integrated task and path planning approach for intelligent coordination of agricultural robots. *Expert Syst. Appl.* **2026**, *302*, 130602. [CrossRef]
132. Dembani, R.; Karvelas, I.; Akbar, N.A.; Rizou, S.; Tegolo, D.; Fountas, S. Agricultural data privacy and federated learning: A review of challenges and opportunities. *Comput. Electron. Agric.* **2025**, *232*, 110048. [CrossRef]
133. Campoverde-Molina, M.; Luján-Mora, S. Cybersecurity in smart agriculture: A systematic literature review. *Comput. Secur.* **2025**, *150*, 104284. [CrossRef]
134. Cantonjos, N.A.P.; Biswas, A. AgroAskAI: A Multi-Agentic AI Framework for Supporting Smallholder Farmers' Enquiries Globally. *Proc. AAAI Conf. Artif. Intell.* **2026**, *40*, 40217–40223. [CrossRef]
135. Ke, Y.; Yu, X.; Du, H.; Chapman, S.; Huang, H. Dynamic Orchestration of Multi-agent System for Real-World Multi-image Agricultural VQA. In *Proceedings of the Australasian Database Conference*; Springer: Singapore, 2025; pp. 153–165. [CrossRef]
136. Li, T.; Xie, F.; Zhao, Z.; Zhao, H.; Guo, X.; Feng, Q. A multi-arm robot system for efficient apple harvesting: Perception, task plan and control. *Comput. Electron. Agric.* **2023**, *211*, 107979. [CrossRef]
137. Xie, F.; Guo, Z.; Li, T.; Feng, Q.; Zhao, C. Dynamic Task Planning for Multi-Arm Harvesting Robots Under Multiple Constraints Using Deep Reinforcement Learning. *Horticulturae* **2025**, *11*, 88. [CrossRef]
138. Guo, Z.; Fu, H.; Wu, J.; Han, W.; Huang, W.; Zheng, W.; Li, T. Dynamic Task Planning for Multi-Arm Apple-Harvesting Robots Using LSTM-PPO Reinforcement Learning Algorithm. *Agriculture* **2025**, *15*, 588. [CrossRef]
139. Gutiérrez Cejudo, J.; Enguix Andrés, F.; Lujak, M.; Carrascosa Casamayor, C.; Fernandez, A.; Hernández López, L. Towards Agrirobot Digital Twins: Agri-RO5—A Multi-Agent Architecture for Dynamic Fleet Simulation. *Electronics* **2024**, *13*, 80. [CrossRef]
140. Chojka, A.; Adão, T.; Pascoal, D.; Silva, N.; Peres, E.; Morais, R. AI Agent-Based Sensor Monitoring for IoT-Enabled Precision Agriculture: A Case Study in Workflow Automation. *Procedia Comput. Sci.* **2026**, *278*, 700–708. [CrossRef]
141. Cao, R.; Li, S.; Ji, Y.; Zhang, Z.; Xu, H.; Zhang, M.; Li, M.; Li, H. Task assignment of multiple agricultural machinery cooperation based on improved ant colony algorithm. *Comput. Electron. Agric.* **2021**, *182*, 105993. [CrossRef]
142. Subeesh, A.; Chauhan, N. Agricultural digital twin for smart farming: A review. *Green Technol. Sustain.* **2026**, *4*, 100299. [CrossRef]
143. Santilli, M.; Carpio, R.F.; Gasparri, A. A Framework for Tasks Allocation and Scheduling in Precision Agriculture Settings. In *Proceedings of the 2021 20th International Conference on Advanced Robotics (ICAR)*; IEEE: New York, NY, USA, 2021; pp. 996–1002. [CrossRef]
144. Xiong, G.; Guo, J.; Parsons, K.; Nagai, Y.; Sumi, T.; Orlik, P.; Li, J. UAV Aided Smart Agriculture Networks: A Multi-Agent Reinforcement Learning Approach. In *Proceedings of the ICC 2025—IEEE International Conference on Communications*; IEEE: New York, NY, USA, 2025; pp. 3075–3081. [CrossRef]
145. Bhat, M.H.; da Silva Rickiel, F.; Sameer, B.; Aeshna, S.; Moore, K.J. Precision Agriculture Through a Real-Time Systems Perspective: A Narrative Review. *Agronomy* **2026**, *16*, 552. [CrossRef]
146. Tsaousidis, M.; Kalampokas, T.; Vrochidou, E.; Papakostas, G.A. AI-Enabled Digital Twins in Agriculture. *AI* **2026**, *7*, 108. [CrossRef]
147. Li, H.; Wu, H.; Li, Q.; Zhao, C. A review on enhancing agricultural intelligence with large language models. *Artif. Intell. Agric.* **2025**, *15*, 671–685. [CrossRef]
148. Zaiwen, F.; Duo, X.; Fang, T.; Hongyu, Z.; Wanli, L.; Hui, P.; Shanmei, L.; Hanzun, L.; Huidong, J.; Yuan, H.; et al. Key technologies and development trends of intelligent decision-making large models for facility agriculture. *Trans. Chin. Soc. Agric. Eng.* **2025**, *41*, 50–61. [CrossRef]

149. Grosse, M.; Honda, K.; Thies, P.; Specht, C. Discrete Event Simulation Based on a Multi-Agent System for Japanese Rice Harvesting Operations. *Agriculture* **2025**, *15*, 1745. [CrossRef]
150. Kitouni, I.; Benmerzoug, D.; Lezzar, F. Smart Agricultural Enterprise System Based on Integration of Internet of Things and Agent Technology. *J. Organ. End User Comput.* **2018**, *30*, 64–82. [CrossRef]
151. Díaz, J.; Quiñonez, Y.; De-la Hoz-Franco, E.; Butt-Aziz, S.; Mercado, T.; Salcedo, D. Information and Communication Technologies Used in Precision Agriculture: A Systematic Review. *AgriEngineering* **2025**, *7*, 167. [CrossRef]
152. Ragu, N.; Teo, J. Object Detection and Classification Using Few-Shot Learning in Smart Agriculture: A Scoping Mini Review. *Front. Sustain. Food Syst.* **2023**, *6*, 1039299. [CrossRef]
153. Zhang, D.; Pan, F.; Diao, Q.; Feng, X.; Li, W.; Wang, J. Seeding Crop Detection Framework Using Prototypical Network Method in UAV Images. *Agriculture* **2021**, *12*, 26. [CrossRef]
154. Gong, R.; Zhang, H.; Li, G.; He, J. Edge Computing-Enabled Smart Agriculture: Technical Architectures, Practical Evolution, and Bottleneck Breakthroughs. *Sensors* **2025**, *25*, 5302. [CrossRef]
155. Omotayo, A.O.; Adediran, S.A.; Omotoso, A.B.; Olagunju, K.O.; Omotayo, O.P. Artificial intelligence in agriculture: Ethics, impact possibilities, and pathways for policy. *Comput. Electron. Agric.* **2025**, *239*, 110927. [CrossRef]
156. Barbedo, J.G.A.; da Silva, M.S.; Cazzolato, M.T.; Valem, L.P.; Tinós, R.; Romero, R.A.F.; Murta, L.O.; de Jesus Holanda, A.; Felipe, J.C.; Pinheiro, J.B.; et al. Explainability and privacy in AI-enabled crop monitoring: Trends and future directions in soybean research. *Comput. Electron. Agric.* **2026**, *243*, 111392. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.